

COMPLETE CALABI-YAU METRICS ON NONCOMPACT ABELIAN FIBERED THREEFOLDS

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ABSTRACT. In this article, we construct new families of complete Calabi-Yau metrics on abelian fibrations X over $\mathbb{C}$, which generalizes the type ALG, ALG*, ALH, ALH* gravitational instantons to higher dimensions. We also provide a projective compactification for X so that the compactified variety has negative canonical divisor.

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1. INTRODUCTION

1.1. Motivation and overview. Following Yau's seminal work on Ricci curvature [24], the study of Calabi-Yau metrics has become a central topic in Kähler geometry. A significant challenge in this field is constructing complete Calabi-Yau metrics on non-compact Kähler manifolds. The systematic study of this topic was initiated by Tian and Yau in [20], focusing primarily on the construction of such metrics on a Fano manifold minus a smooth anti-canonical divisor. They also addressed the case of a non-Fano Calabi-Yau fibration with a smooth fiber removed.

A primary setting since then has been the complex two-dimensional case, specifically the construction and classification of so-called gravitational instantons. A gravitational instanton is defined as a non-compact, complete, non-flat hyperkähler 4-manifold M satisfying $|\text{Rm}(x)| < C \text{dist}(x, x_0)^{-2}$ as $x \rightarrow \infty$. There have been numerous works on the construction and classification of gravitational instantons with prescribed asymptotics at infinity (see, for example, [1, 2, 3, 4, 8, 14, 15, 19]).

Gravitational instantons are classified according to their asymptotic behavior at infinity into the types ALE, ALF, ALG, ALH, ALG*, and ALH*. Building on the framework of [20], Hein introduced a novel method for constructing complete Calabi-Yau metrics on rational elliptic surfaces with certain singular fibers removed. Hein's constructions [8] provide numerous examples of types ALG, ALH, ALG*, and ALH*.

Inspired by Hein's work, we construct higher-dimensional analogues of gravitational instantons of types ALG, ALH, ALG*, and ALH*. More precisely, we construct complete noncompact Calabi-Yau metrics on abelian fibrations with prescribed asymptotic models at infinity and quadratic curvature decay with respect to the distance function. These constructions are genuinely higher-dimensional: they cannot be recovered from known lower-dimensional examples by taking products with flat tori and passing to finite quotients, as made precise in Remark 1.4.

1.2. Main results.

Definition 1.1 (Abelian-fibered threefold). *By an abelian-fibered threefold, we mean a normal three-dimensional variety X equipped with a projective surjective morphism $f : X \rightarrow C$ to a normal curve C whose general fiber is an abelian surface.*

Definition 1.2 (Rational-elliptic type). *A three-dimensional variety X is said to be of rational-elliptic type if there exist a smooth rational elliptic surface $g : S \rightarrow \mathbb{P}^1$, an elliptic curve E , a finite subgroup $\Gamma \subseteq E$, and a homomorphism $\rho : \Gamma \rightarrow \text{MW}(S)$ such that*

$$X \cong (S \times E)/\Gamma,$$

where the action is given by

$$\gamma \cdot (s, a) = (\rho(\gamma) \cdot s, a + \gamma).$$

We now state the main result of the paper. The two constructions underlying the theorem will be presented in detail in Sections 4 and 5.

Theorem 1.3 (Main theorem). *There exist abelian-fibered threefolds X equipped with relatively minimal surjective morphisms $f : X \rightarrow \mathbb{P}^1$ in the sense of Definition 1.6, satisfying the following conditions:*

- (1) *The singular locus of X is contained in the fiber $f^{-1}(\infty)$;*
- (2) *The canonical divisor K_X is $\mathbb{Q}$ -linearly equivalent to $-\lambda f^*(\infty)$ for some $\lambda \in \mathbb{Q}_{>0}$;*
- (3) *Writing $M = X \setminus f^{-1}(\infty)$, there **do not** exist a threefold Y of rational-elliptic type, its induced abelian fibration $g : Y \rightarrow \mathbb{P}^1$, and a point $q \in \mathbb{P}^1$ such that*

$$M \cong Y \setminus g^{-1}(q)$$

as complex manifolds. Here the biholomorphism is not assumed a priori to preserve the fibrations;

- (4) *The open manifold M admits a complete Calabi-Yau metric of generalized ALG, ALH, ALG*, or ALH* type, in the sense of Definitions 3.1 and 3.6. Its asymptotic type is determined by the local monodromy around the deleted fiber $f^{-1}(\infty)$.*

For the proof of Theorem 1.3, we refer the reader to the end of Subsection 7.1.

Remark 1.4. *Condition (3) expresses the precise sense in which the metrics constructed in this paper are nontrivial. Indeed, let $Y = (S \times E)/\Gamma$ be a threefold of rational-elliptic type. The diagonal action of Γ is free. Fix a point q in $\mathbb{P}^1$, after choosing the data in Hein's construction to be $\rho(\Gamma)$ -invariant, one obtains a $\rho(\Gamma)$ -invariant gravitational instanton metric on $S \setminus g^{-1}(q)$. Its product with a translation-invariant flat metric on E is a Γ -invariant Calabi-Yau metric and therefore descends to the quotient. This construction is essentially the product of a four-dimensional gravitational instanton with a flat elliptic curve. Condition (3) rules out precisely this product-quotient construction: the open Calabi-Yau threefolds constructed here are not even biholomorphic to the*

complement of a fiber in a threefold of rational-elliptic type, see Proposition 7.4 and Proposition 7.5.

Remark 1.5. We give two complementary constructions. The first is based on isotrivial quotient models, while the second uses non-isotrivial fiber products of rational elliptic surfaces; see Theorems 4.1 and 5.1, respectively. In these constructions, the asymptotic type is determined by the local monodromy around the deleted fiber. Roughly speaking, finite monodromy gives rise to generalized ALG or ALH geometry, whereas the presence of a nontrivial unipotent part leads to the corresponding starred types. The precise relation between the monodromy and the resulting asymptotic model is established in the constructions in Theorems 4.1 and 5.1.

1.3. Strategy and main obstructions. The main strategy for constructing the Calabi–Yau metric consists of the following steps:

- (1) Find an appropriate geometric model satisfying the first three conditions in Theorem 1.3;
- (2) Construct an explicit ansatz near infinity;
- (3) Prove a $\partial\bar{\partial}$ -type lemma and extend the ansatz to the whole space by a gluing argument;
- (4) Solve the complex Monge–Ampère equation to perturb the resulting ansatz into a Calabi–Yau metric.

The additional difficulties in higher dimensions are concentrated primarily in the first and third steps. The first is algebro-geometric: one must construct a suitable compactified fibration with sufficiently mild singularities and a controlled canonical divisor. The third is topological: one must ensure that the semi-flat Kähler class near infinity is compatible with the restriction of a global Kähler class.

In Hein’s work, the principal geometric object is a rational elliptic surface X with a distinguished fiber F removed. Recall that a rational elliptic surface is a smooth surface equipped with a relatively minimal elliptic fibration over $\mathbb{P}^1$ such that $K_X \sim -F$. This relation is essential to the construction and is closely connected with Kodaira’s theory of smooth relatively minimal elliptic surfaces [13]. In higher dimensions, however, the situation is substantially more complicated, largely because of the subtleties of the minimal model program.

Definition 1.6 (Relatively minimal model). *An abelian-fibered threefold $f : X \rightarrow S$ is called relatively minimal if*

- (1) *X has at worst klt singularities;*
- (2) *The canonical Weil divisor K_X is relatively nef, that is, for every curve C contained in a fiber X_s , one has $K_X \cdot C \geq 0$.*

Remark 1.7. *The convention in Definition 1.6 is weaker than the corresponding notions used by Oguiso [16] and by Crauder–Morrison [5]. In particular, we allow the ambient threefold to have nonterminal klt singularities.*

The key feature of a relatively minimal fibration over $\mathbb{P}^1$ is that the canonical divisor of the ambient variety is $\mathbb{Q}$ -linearly equivalent to a multiple of the fiber class.

Lemma 1.8. *Let $f : X \rightarrow \mathbb{P}^1$ be a relatively minimal abelian-fibered threefold. Then there exists $\lambda \in \mathbb{Q}$ such that $K_X \sim_{\mathbb{Q}} \lambda F$, where $F = f^*(p)$ is the scheme-theoretic fiber over any point $p \in \mathbb{P}^1$.*

Proof. By the relative log abundance theorem for threefolds [7, Theorem A.2], the f -nef divisor K_X is f -semiample. Choose $m > 0$ sufficiently divisible so that mK_X is Cartier and f -generated. Since the generic fiber A_η is an abelian surface, $mK_X|_{A_\eta} \sim 0$. Consequently, $f_*\mathcal{O}_X(mK_X)$ is a torsion-free sheaf of rank one on $\mathbb{P}^1$, hence a line bundle. Thus we have $f^*f_*\mathcal{O}_X(mK_X) \cong \mathcal{O}_X(mK_X)$, and we obtain $mK_X \sim f^*L \sim dF$ for some d . Therefore, $K_X \sim_{\mathbb{Q}} \frac{d}{m}F$. $\square$

In contrast with the surface case, finding a nontrivial higher dimensional abelian fibration satisfying the first three conditions of Theorem 1.3 is a central difficulty. Even when a relatively minimal

model $X_{\min}$ exists, it need not be smooth and its singular locus need not be contained in one single fiber, and the model need not be unique. These complications also prevent one from expecting a classification as uniform and explicit as the classification of rational elliptic surfaces.

On the other hand, genuinely new topological obstructions arise in the gluing step when the fiber dimension increases. For an elliptic curve, the group $H^2(F, \mathbb{Z})$ has rank one, so its Kähler class is determined by its total area. If F is a complex torus of dimension $g \geq 2$, however, then $\text{rank } H^2(F, \mathbb{Z}) = \binom{2g}{2} > 1$; in particular, an abelian surface has second Betti number 6. It is therefore not enough to match only the total volume of the fibers. To carry out the gluing construction, one must find a global Kähler form whose restriction to every smooth fiber represents the same cohomology class as the semi-flat form, or equivalently, whose periods agree with those of the semi-flat form on all relevant monodromy-invariant two-cycles. Failure of these period-matching conditions obstructs the required $\partial\bar{\partial}$ -correction. Thus the construction of the global ansatz involves genuine topological constraints and is considerably more delicate than in the elliptic-surface case.

1.4. Organization of the paper. The paper is organized as follows.

In Section 2, we review the construction of semi-flat Ricci-flat metrics on smooth torus fibrations and fix the conventions used in the subsequent explicit calculations.

In Section 3, we introduce the higher-dimensional asymptotic model geometries used throughout the paper. We define the generalized ALG, ALH, ALG^* , and ALH^* model spaces, together with the corresponding asymptotic classes of Calabi–Yau metrics.

In Section 4, we study isotrivial quotient models, analyze their singularities and canonical divisors, and describe the associated generalized ALG models. These examples also illustrate several features of the more involved non-isotrivial construction considered in the following section.

In Section 5, we construct projective compactifications of non-isotrivial abelian-surface fibrations arising from fiber products of rational elliptic surfaces. We then construct explicit semi-flat ansatz metrics near infinity and determine their asymptotic behavior from the local monodromy of the distinguished fiber.

In Section 6, we formulate the cohomological matching condition required for the relevant $\partial\bar{\partial}$ -lemma and use it to extend the semi-flat ansatz to a global Kähler metric. We then apply the Tian–Yau–Hein package and establish the asymptotic comparison between the resulting Calabi–Yau metric and the corresponding model metric.

In Section 7, we establish the nontriviality of our constructions. We prove that the isotrivial quotient models and non-isotrivial fiber-product models, after deletion of the distinguished fiber, are not biholomorphic to the complement of a fiber in a threefold of rational-elliptic type. We also show that the generalized ALH_{b_1, b_2}^* metrics arising from the $I_{b_1}^* \times I_{b_2}^*$ construction cannot be identified with three-dimensional Tian–Yau metrics, despite having the same volume-growth order.

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2. SEMI-FLAT METRICS ON TORUS FIBRATIONS

We collect the semi-flat construction for a holomorphic family of polarized complex tori over a Riemann surface and fix the period-matrix conventions used below. After describing the Gauss–Manin horizontal distribution, we derive the local expression for the semi-flat Calabi–Yau form. We then specialize this formula to fiber products of elliptic fibrations.

Let $f : X \rightarrow S$ be a holomorphic submersion over a Riemann surface S , not necessarily compact, whose fibers $X_s = f^{-1}(s)$ are complex m -tori. Assume that f admits a holomorphic section $\sigma : S \rightarrow X$ and that $\omega \in \Omega^2(X, \mathbb{R})$ represents a constant polarization. More precisely, each

restriction $\omega|_{X_s}$ is a Kähler form, and locally on S one can write $[\omega|_{X_s}] = \sum_i c_i \eta_i(s)$, where $c_i \in \mathbb{R}$ are constant and the classes $\eta_i(s) \in H^2(X_s, \mathbb{Z})$ are Gauss–Manin parallel.

The following observation shows that any closed real two-form whose restrictions to the fibers are Kähler automatically determines a constant polarization.

Lemma 2.1. [10, Proposition 2.1] *Suppose that $X_s \cong \mathbb{C}^m/\Lambda_s$. Fix a basis $(\tau_1(s), \dots, \tau_{2m}(s))$ of the lattice Λ_s that varies holomorphically with $s \in S$, and let $(\xi^1(s), \dots, \xi^{2m}(s))$ be the $\mathbb{R}$ -dual basis of real one-forms on $\mathbb{C}^m$. Then the classes $[\xi^i(s) \wedge \xi^j(s)] \in H^2(X_s, \mathbb{Z}) \subset H^2(X_s, \mathbb{R})$, $i < j$, form a basis of $H^2(X_s, \mathbb{Z})$.*

If a real two-form ω on X is closed and $[\omega|_{X_s}] = \sum_{i < j} P_{ij}(s) [\xi^i(s) \wedge \xi^j(s)]$, then the functions P_{ij} are locally constant.

For $\varepsilon > 0$ and $s \in S$, let $g_{s,\varepsilon}$ be the unique translation-invariant flat Kähler metric on X_s whose Kähler class is proportional to $[\omega|_{X_s}]$ and whose volume is ε . Equivalently, if $\omega_{s,\varepsilon}$ denotes its Kähler form, then $[\omega_{s,\varepsilon}] = c_{s,\varepsilon}[\omega|_{X_s}]$ for some $c_{s,\varepsilon} > 0$, determined uniquely by the normalization $\frac{1}{m!} \int_{X_s} \omega_{s,\varepsilon}^m = \varepsilon$. Restricting $g_{s,\varepsilon}$ to $T_{\sigma(s)}X_s$ induces a Hermitian metric h_ε on the holomorphic vector bundle $E := \sigma^*T_{X/S}$. More precisely,

$$h_{\varepsilon,s}(v, w) := g_{s,\varepsilon}(v, \bar{w}), \quad v, w \in T_{\sigma(s)}^{1,0}X_s = E_s.$$

Assume further that X admits a nowhere-vanishing holomorphic volume form Ω . Then Ω induces a nondegenerate pairing

$$\bigwedge^m E \otimes T^{1,0}S \longrightarrow \mathbb{C}, \quad ((v_1, \dots, v_m), w) \longmapsto \Omega(v_1, \dots, v_m, \sigma_* w).$$

Using the Hermitian metric on $\bigwedge^m E$ induced by h_ε , we obtain a Hermitian metric $g_{S,\varepsilon}$ on S . Explicitly, for any basis $v_1, \dots, v_m$ of E_s and any $w \in T_s^{1,0}S$, it is given by

$$g_{S,\varepsilon}(w, \bar{w}) = \frac{|\Omega(v_1, \dots, v_m, \sigma_* w)|^2}{\det(h_{\varepsilon,s}(v_i, v_j))}.$$

Using $g_{S,\varepsilon}$, the family of flat metrics $\{g_{s,\varepsilon}\}$ on the fibers, and the horizontal distribution described below, we can construct a Riemannian submersion metric on X .

Lemma 2.2. *Over a simply connected coordinate neighborhood $U \subset S$, choose a marking of the homology local system $\Lambda_{\mathbb{Z}} := (R^1 f_* \mathbb{Z})^\vee$ and identify $X|_U \cong E|_U/\Lambda$, where Λ denotes the corresponding lattice in the underlying real vector bundle $E_{\mathbb{R}}|_U$.*

- (1) *The lattice Λ induces a flat $\mathbb{R}$ -linear connection on $E_{\mathbb{R}}|_U$, and hence an integrable horizontal distribution $\mathcal{H}$ on $X|_U$.*
- (2) *The horizontal distribution $\mathcal{H}$ is independent of the choice of the marking and of the corresponding biholomorphism*

$$X|_U \cong E|_U/\Lambda.$$

Proof. (1) Let $\lambda_1, \dots, \lambda_{2m}$ be the local frame of $E_{\mathbb{R}}|_U$ determined by a basis of the lattice Λ . Every smooth section u of $E_{\mathbb{R}}|_U$ can be written uniquely as $u = \sum_{i=1}^{2m} a_i \lambda_i$ for smooth real-valued functions a_i on U . Define $\nabla u = \sum_{i=1}^{2m} \lambda_i \otimes da_i$. This is a flat $\mathbb{R}$ -linear connection for which the lattice sections λ_i are parallel. Since the connection preserves Λ , its horizontal distribution descends from $E_{\mathbb{R}}|_U$ to the quotient $E|_U/\Lambda \cong X|_U$. Flatness of ∇ implies that the resulting horizontal distribution $\mathcal{H}$ is integrable.

(2) Two markings of the homology local system differ by a locally constant transformation $A \in \mathrm{GL}(2m, \mathbb{Z})$. Consequently, the corresponding lattice frames are related by a locally constant change of basis. The definition of ∇ is invariant under such a change of frame, and hence the induced horizontal distribution $\mathcal{H}$ is independent of the chosen marking and uniformization. $\square$

Define

$$g_{\text{sf},\varepsilon}(u, v) := g_{S,\varepsilon}(f_*u, f_*v) + g_{s,\varepsilon}(Pu, Pv), \quad u, v \in T_x X, \quad s = f(x),$$

where $P : T_x X \rightarrow \mathcal{V}_x = T_x X_s$ is the projection along $\mathcal{H}_x$. This defines a semi-flat Hermitian metric $g_{\text{sf},\varepsilon}$ on X .

For computational purposes, we now derive an explicit local formula for $g_{\text{sf},\varepsilon}$ in terms of Ω and the periods of the tori X_s with respect to ω .

Fix isomorphisms $X|_U \cong E|_U/\Lambda \cong (U \times \mathbb{C}^m)/\Lambda$ over an open set $U \subset \mathbb{C}$, with coordinate z on U and coordinates $v^1, \dots, v^m$ along the fibers. Fix an oriented basis $(\tau_1, \dots, \tau_{2m})$ of Λ at one point and extend it by analytic continuation to a tuple of possibly multivalued holomorphic functions $\tau_i(z) \in \mathcal{O}(U, \mathbb{C}^m)$ that generates Λ everywhere. Let $(\xi^1(z), \dots, \xi^{2m}(z))$ be the $\mathbb{R}$ -dual basis of $(\tau_1(z), \dots, \tau_{2m}(z))$. We now relate the complex coordinates $v^1, \dots, v^m$ to this real basis. Write

$$\tau_i(z) = (T_{1,i}(z), \dots, T_{m,i}(z))^{\text{tr}}, \quad i = 1, \dots, 2m,$$

and define

$$T = \begin{pmatrix} T_{1,1} & \cdots & T_{1,2m} \\ \vdots & \ddots & \vdots \\ T_{m,1} & \cdots & T_{m,2m} \end{pmatrix} = (\tau_1 \cdots \tau_{2m}) \in \mathcal{O}(U, \mathbb{C}^{m \times 2m}).$$

Then

$$(dv^1, \dots, dv^m) = (\xi^1, \dots, \xi^{2m}) \begin{pmatrix} T_{1,1} & \cdots & T_{m,1} \\ \vdots & \ddots & \vdots \\ T_{1,2m} & \cdots & T_{m,2m} \end{pmatrix}.$$

Equivalently,

$$\begin{pmatrix} dv \\ d\bar{v} \end{pmatrix} = \begin{pmatrix} T \\ \bar{T} \end{pmatrix} \xi, \quad \xi = (\Pi, \bar{\Pi}) \begin{pmatrix} dv \\ d\bar{v} \end{pmatrix},$$

where $\Pi \in \mathcal{C}^\infty(U, \mathbb{C}^{2m \times m})$ satisfies $(\Pi, \bar{\Pi}) \begin{pmatrix} T \\ \bar{T} \end{pmatrix} = I_{2m}$.

The existence of a constant polarization is equivalent to the existence of a real skew-symmetric matrix $Q \in \mathbb{R}^{2m \times 2m}$ such that $\omega = \frac{1}{2} \sum_{i,j} Q_{ij} \xi^i \wedge \xi^j$ restricts to a flat Kähler form on each fiber.

The following standard linear-algebraic facts concerning polarized torus fibrations are contained, in equivalent notation, in Hein's discussion of semi-flat metrics; see [8, §5.3, equations (5.1)–(5.2)]. We record them in our notation and omit the proof.

Lemma 2.3. (1) Under a monodromy transformation, T transforms as $T \mapsto TA$, where $A \in \text{GL}(2m, \mathbb{Z})$ satisfies $A^{\text{tr}}QA = Q$.

(2) There exists $S \in \text{GL}(2m, \mathbb{R})$ such that $S^{\text{tr}}QS = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}$.

(3) Write $TS = R(I_m, Z)$, where $R : U \rightarrow \text{GL}(m, \mathbb{C})$, $Z : U \rightarrow \mathbb{C}^{m \times m}$ are possibly multivalued holomorphic maps. Then ω is a positive $(1, 1)$ -form if and only if $Z \in \mathfrak{H}_m$, where $\mathfrak{H}_m$ is the Siegel upper half-space.

(4) One has $\omega = iH_{jk} dv^j \wedge d\bar{v}^k$, where $H^{-1} := 2\bar{R}(\text{Im } Z)R^{\text{tr}} = i\bar{T}Q^{-1}T^{\text{tr}}$.

Since the one-forms $\xi^1, \dots, \xi^{2m}$ are parallel with respect to the flat connection induced by Λ , one has

$$\nabla_{\partial/\partial z} \begin{pmatrix} dv^1 \\ \vdots \\ dv^m \end{pmatrix} = \frac{\partial T}{\partial z} \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^{2m} \end{pmatrix} = \frac{\partial T}{\partial z} \begin{pmatrix} T \\ \bar{T} \end{pmatrix}^{-1} \begin{pmatrix} dv \\ d\bar{v} \end{pmatrix}.$$

Therefore, the flat connection induced by Λ has Christoffel symbol

$$\Gamma(z, v) = \frac{\partial T}{\partial z} \begin{pmatrix} T \\ \bar{T} \end{pmatrix}^{-1} \begin{pmatrix} v \\ \bar{v} \end{pmatrix} \in \mathbb{C}^m.$$

Equivalently, the vectors $\left(\frac{\partial}{\partial z}, \sum_{j=1}^m \Gamma^j(z, v) \frac{\partial}{\partial v^j}\right)$ and their complex conjugates span the horizontal space at $(z, v) \in U \times \mathbb{C}^m$.

Theorem 2.4. [8, Lemma 5.3] Define $H(\varepsilon) := \left(\frac{\varepsilon}{\sqrt{\det Q}}\right)^{1/m} H$, where $H^{-1} := 2\bar{R}(\text{Im } Z)R^{\text{tr}} = i\bar{T}Q^{-1}T^{\text{tr}}$. If $\Omega = g dz \wedge dv^1 \wedge \cdots \wedge dv^m$ for some holomorphic function $g : U \rightarrow \mathbb{C}$, then the Kähler form of the Hermitian metric $g_{\text{sf}, \varepsilon}$ is

$$\omega_{\text{sf}, \varepsilon} = i|g|^2 \det(H(\varepsilon))^{-1} dz \wedge d\bar{z} + iH(\varepsilon)_{jk} (dv^j - \Gamma^j dz) \wedge (d\bar{v}^k - \bar{\Gamma}^k d\bar{z}).$$

This form is closed and satisfies

$$\omega_{\text{sf}, \varepsilon}^{m+1} = (m+1)! i^{(m+1)^2} \Omega \wedge \bar{\Omega}.$$

In particular, $g_{\text{sf}, \varepsilon}$ is a Calabi–Yau metric.

For a fiber product of two elliptic fibrations equipped with the product polarization, both the period matrix and the fiber metric split into two elliptic blocks. Direct substitution into the formula of Theorem 2.4 gives the following corollary; we omit the straightforward calculation.

Corollary 2.5. Let $f : X \rightarrow S$ be the fiber product of two smooth elliptic fibrations over a Riemann surface S , equipped with a holomorphic section σ . Let Ω be a holomorphic volume form on X , and let $\omega_{\text{sf}, \varepsilon}$ denote the semi-flat Calabi–Yau metric obtained from the product polarization with total abelian-surface fiber volume ε^2 , so that the restriction to each elliptic factor has area ε . Then

$$\omega_{\text{sf}, \varepsilon}^3 = 6i\Omega \wedge \bar{\Omega}.$$

Let U be a domain in S , let z be a holomorphic coordinate on U , and fix an isomorphism of elliptic fibrations

$$X|_U \cong (U \times \mathbb{C}_{v_1} \times \mathbb{C}_{v_2}) / \Lambda,$$

where Λ is generated by $(\tau_1, 0)$, $(\tau_2, 0)$, $(0, \tau_3)$, $(0, \tau_4)$, for possibly multivalued holomorphic functions $\tau_1, \tau_2, \tau_3, \tau_4 : U \rightarrow \mathbb{C}$, such that the corresponding lattice basis is positively oriented and σ maps to the zero section. Then $\Omega = g dz \wedge dv_1 \wedge dv_2$ for some holomorphic function $g : U \rightarrow \mathbb{C}$, and

$$\begin{aligned} \omega_{\text{sf}, \varepsilon} &= 4i|g|^2 \frac{\text{Im}(\bar{\tau}_1 \tau_2) \text{Im}(\bar{\tau}_3 \tau_4)}{\varepsilon^2} dz \wedge d\bar{z} \\ &\quad + \frac{i\varepsilon}{2 \text{Im}(\bar{\tau}_1 \tau_2)} (dv_1 - \Gamma^1 dz) \wedge (d\bar{v}_1 - \bar{\Gamma}^1 d\bar{z}) \\ &\quad + \frac{i\varepsilon}{2 \text{Im}(\bar{\tau}_3 \tau_4)} (dv_2 - \Gamma^2 dz) \wedge (d\bar{v}_2 - \bar{\Gamma}^2 d\bar{z}), \end{aligned} \tag{2.1}$$

where

$$\Gamma^1(z, v) = \frac{\text{Im}(\bar{\tau}_1 v_1) \tau_2' - \text{Im}(\bar{\tau}_2 v_1) \tau_1'}{\text{Im}(\bar{\tau}_1 \tau_2)}, \quad \Gamma^2(z, v) = \frac{\text{Im}(\bar{\tau}_3 v_2) \tau_4' - \text{Im}(\bar{\tau}_4 v_2) \tau_3'}{\text{Im}(\bar{\tau}_3 \tau_4)}.$$

3. ASYMPTOTIC MODEL GEOMETRIES

This section fixes the asymptotic models used to state the metric results. We first recall product-type higher-dimensional ALG and ALH ends. We then define semi-flat abelian-surface ends associated with doubly unipotent and mixed unipotent–finite monodromy, which will be called generalized ALH* and generalized ALG* models, respectively.

Definition 3.1 (Generalized ALH, ALG metrics). Let N be a noncompact manifold, and let g be a complete Riemannian metric on N .

(a) Fix $\beta \in (0, 1]$ and an isometry γ of a flat torus (T^{2m}, g_T) . Let

$$E_{\beta, \gamma}(R) = ((R, \infty) \times [0, 2\pi\beta] \times T^{2m}) / \sim,$$

where $(r, 0, x) \sim (r, 2\pi\beta, \gamma(x))$, and equip it with the metric

$$g_{\text{flat}} = dr^2 + r^2 d\vartheta^2 + g_T.$$

We call g a generalized ALG metric of cone angle $2\pi\beta$ if there exist a compact set $K \subset N$, constants $R, \delta > 0$, and a diffeomorphism

$$\Phi : E_{\beta, \gamma}(R) \longrightarrow N \setminus K$$

such that

$$\left| \nabla_{g_{\text{flat}}}^k (\Phi^* g - g_{\text{flat}}) \right|_{g_{\text{flat}}} \leq C_k r^{-\delta-k}, \quad k \in \mathbb{Z}_{\geq 0}.$$

(b) Fix $L > 0$ and an isometry γ of a flat torus (T^{2m}, g_T) , and set

$$Y_{L, \gamma} = ([0, L] \times T^{2m}) / \sim, \quad (0, x) \sim (L, \gamma(x)).$$

The metric $d\vartheta^2 + g_T$ descends to a flat metric h on $Y_{L, \gamma}$. We call g a generalized ALH metric if there exist a compact set $K \subset N$, constants $R, \delta > 0$, and a diffeomorphism

$$\Phi : (R, \infty) \times Y_{L, \gamma} \longrightarrow N \setminus K$$

such that

$$\left| \nabla_{g_{\text{flat}}}^k (\Phi^* g - g_{\text{flat}}) \right|_{g_{\text{flat}}} \leq C_k e^{-\delta t}, \quad k \in \mathbb{Z}_{\geq 0},$$

where $g_{\text{flat}} = dt^2 + h$. When $\gamma = \text{id}$, the cross-section $Y_{L, \gamma}$ is a flat $(2m+1)$ -torus.

Remark 3.2. These definitions directly generalize those in Hein's thesis. Their tangent cones at infinity are, respectively, a flat two-dimensional cone and a ray. Replacing the torus factor by another compact Calabi–Yau manifold leads to a different class of model ends.

Definition 3.3 (Semi-flat abelian-surface model end). Let

$$\Delta^* = \{z \in \mathbb{C} : 0 < |z| < 1\}, \quad \mathcal{E} = (\Delta^* \times \mathbb{C}^2) / \Lambda(z),$$

where $\Lambda(z)$ is generated by $(\tau_1(z), 0)$, $(\tau_2(z), 0)$, $(0, \tau_3(z))$, $(0, \tau_4(z))$. The period functions are understood locally on the universal cover of Δ^* , and their analytic continuations are assumed to preserve the lattice.

Set

$$\Gamma^1 = \frac{\text{Im}(\bar{\tau}_1 v_1) \tau_2' - \text{Im}(\bar{\tau}_2 v_1) \tau_1'}{\text{Im}(\bar{\tau}_1 \tau_2)}, \quad \Gamma^2 = \frac{\text{Im}(\bar{\tau}_3 v_2) \tau_4' - \text{Im}(\bar{\tau}_4 v_2) \tau_3'}{\text{Im}(\bar{\tau}_3 \tau_4)}.$$

For $\varepsilon > 0$, $\kappa \in \mathbb{C}^*$, and

$$\Omega_0 = \kappa z^{-2} dz \wedge dv_1 \wedge dv_2,$$

define

$$\begin{aligned} \omega_0 &= \frac{4i|\kappa|^2 \text{Im}(\bar{\tau}_1 \tau_2) \text{Im}(\bar{\tau}_3 \tau_4)}{\varepsilon^2 |z|^4} dz \wedge d\bar{z} \\ &\quad + \frac{i\varepsilon}{2 \text{Im}(\bar{\tau}_1 \tau_2)} (dv_1 - \Gamma^1 dz) \wedge (d\bar{v}_1 - \bar{\Gamma}^1 d\bar{z}) \\ &\quad + \frac{i\varepsilon}{2 \text{Im}(\bar{\tau}_3 \tau_4)} (dv_2 - \Gamma^2 dz) \wedge (d\bar{v}_2 - \bar{\Gamma}^2 d\bar{z}). \end{aligned}$$

We denote the associated Riemannian metric by g_0 and set

$$r(x) = 1 + \text{dist}_{g_0}(x, \Sigma),$$

where Σ is a fixed compact cross-section of the end.

Definition 3.4 (Generalized ALH* models). *For $b_1, b_2 \in \mathbb{Z}_{>0}$, the generalized ALH_{b_1, b_2}^* model end is the preceding semi-flat end (Definition 3.3) with*

$$\tau_1 = z^{1/2}, \quad \tau_2 = \frac{b_1}{2\pi i} z^{1/2} \log z, \quad \tau_3 = z^{1/2}, \quad \tau_4 = \frac{b_2}{2\pi i} z^{1/2} \log z.$$

Definition 3.5 (Generalized ALG* models). *Fix $b \in \mathbb{Z}_{>0}$ and $T \in \{\text{II}^*, \text{III}^*, \text{IV}^*\}$. The generalized $\text{ALG}_{b, T}^*$ model end is the preceding semi-flat end with*

$$\tau_1 = z^{1/2}, \quad \tau_2 = \frac{b}{2\pi i} z^{1/2} \log z,$$

and, respectively,

T	τ_3	τ_4
II^*	$z^{1/6}$	$e^{2\pi i/3} z^{1/6}$
III^*	$z^{1/4}$	$iz^{1/4}$
IV^*	$z^{1/3}$	$e^{2\pi i/3} z^{1/3}$

Definition 3.6 (Generalized ALG* and ALH* metrics). *Let (M^6, g) be a complete six-dimensional Riemannian manifold. We call g a generalized $\text{ALG}_{b, T}^*$ metric, respectively a generalized ALH_{b_1, b_2}^* metric, if there exist a compact set $K \subset M$, a truncated end $(\mathcal{E}_R, g_0)$ of the corresponding model, a diffeomorphism*

$$\Phi : \mathcal{E}_R \longrightarrow M \setminus K,$$

and a constant $\delta > 0$ such that, for every $j \in \mathbb{Z}_{\geq 0}$,

$$|\nabla_{g_0}^j (\Phi^* g - g_0)|_{g_0} = O(r^{-j-\delta}).$$

Remark 3.7. *A direct calculation shows that the generalized ALH_{b_1, b_2}^* model has volume growth of order $R^{3/2}$ and tangent cone $\mathbb{R}_+$. The generalized $\text{ALG}_{b, T}^*$ models have quadratic volume growth, and their tangent-cone angles are $\frac{2\pi}{3}$, $\frac{\pi}{2}$, and $\frac{\pi}{3}$ for $T = \text{II}^*$, $T = \text{III}^*$, and $T = \text{IV}^*$, respectively.*

4. CONSTRUCTION OF ALG-TYPE METRICS VIA ISOTRIVIAL MODELS

In this section, we construct complete Calabi–Yau metrics of generalized ALG type on isotrivial abelian fibrations. We begin with a finite quotient of $\mathbb{P}^1 \times A$, where A is an abelian variety, and assume that the quotient singularities over one fixed point of the base admit a projective crepant resolution. The fiber over the other fixed point is retained as an irreducible multiple fiber. We then use the fixed-subspace stratification of quotient singularities and the theory of QALE Kähler metrics to extend the descended flat metric across the exceptional locus.

At the first glance, our construction might look similar to Wang’s construction for generalized ALG Calabi–Yau metrics on crepant resolutions of isotrivial quotients [23, Theorem 1.2]. However, a crucial hypothesis in Wang’s theorem is that the compact Calabi–Yau fiber Y satisfies $b^1(Y) = 0$. This assumption is used in the local $i\partial\bar{\partial}$ -lemma of [23, Section 4] to eliminate the mixed cohomology components and thereby modify a given Kähler metric so that it agrees exactly with the product model outside a compact set. In our setting, however, $b^1(A) = 2 \dim_{\mathbb{C}} A > 0$, so this local $i\partial\bar{\partial}$ -lemma is not applicable.

Instead, we construct the exact isotrivial ansatz directly. The crepant resolution is automatically a local product resolution along the fixed-subspace stratification, and the QALE construction of Joyce [11, Proposition 3.2 and Theorem 4.7] allows the descended flat quotient metric to be extended across the exceptional locus while remaining exactly equal to the flat model outside a compact set. In this sense, Theorem 4.1 complements Wang’s theorem and provides a partial extension of the existence result to isotrivial fibrations whose compact fibers have nonvanishing first Betti number. Once the exact isotrivial ansatz has been constructed, the condition $b^1(A) = 0$ is no longer needed,

and Wang's Schwartz-decay theorem [23, Proposition 1.6] applies to the resulting Tian–Yau–Hein solution.

Theorem 4.1 (Isotrivial construction). *Let A be an abelian variety of complex dimension $d \geq 2$, let $n = d + 1$, and let Ω_A be a nowhere-vanishing holomorphic d -form on A . Suppose that $\tau \in \text{Aut}(A)$ has order k and satisfies $\tau^*\Omega_A = \zeta_k^{-1}\Omega_A$, where $\zeta_k = \exp(\frac{2\pi i}{k})$. Let $G = \langle \gamma \rangle \cong \mathbb{Z}/k\mathbb{Z}$ act on $\mathbb{P}_{[t:s]}^1 \times A$ by $\gamma([t:s], a) = ([\zeta_k t:s], \tau(a))$, and set $Y = (\mathbb{P}^1 \times A)/G$.*

Assume that there exists a projective birational morphism $\pi : X \rightarrow Y$ which is an isomorphism over $Y \setminus Y_0$, where Y_0 is the image of $\{0\} \times A$, and whose restriction over a neighborhood of Y_0 is a crepant resolution. The quotient map $\mathbb{P}^1 \rightarrow \mathbb{P}^1/G \cong \mathbb{P}^1$ induces an abelian fibration $f : X \rightarrow \mathbb{P}^1$. Let $F = f^(\infty)$ be the scheme-theoretic fiber over the image of $t = \infty$, and set $M = X \setminus F$. Then the following statements hold.*

- (1) *The divisor F is irreducible and nonreduced. More precisely,*

$$F = kF_{\text{red}},$$

and F_{red} is naturally identified with $A/\langle \tau \rangle$.

- (2) *The variety X is projective, is smooth on M , and has at worst quotient singularities along F . Moreover,*

$$K_X \sim_{\mathbb{Q}} -2F_{\text{red}} = -\frac{2}{k}F. \quad (4.1)$$

In particular, X has klt singularities and f is relatively minimal. There is a nowhere-vanishing holomorphic n -form Ω on M which extends meromorphically to X with a pole of order two along F_{red} .

- (3) *Fix a G -invariant polarization on A and fix the volume of the flat metric on each smooth fiber. For every sufficiently large $\lambda > 0$, there exist a complete Kähler metric $\omega'_{0,\lambda}$ on M and a smooth function u such that*

$$\omega_{\text{CY}} = \omega'_{0,\lambda} + i\partial\bar{\partial}u$$

is a complete Calabi–Yau metric satisfying

$$\omega_{\text{CY}}^n = n!\lambda i^{n^2} \Omega \wedge \bar{\Omega}.$$

The metric $\omega'_{0,\lambda}$ agrees exactly with the flat isotrivial ALG model outside a compact subset of M . If r denotes the radius function of this model, then, for every $\beta > 0$ and every $j \in \mathbb{N}_0$,

$$\left| \nabla_{\omega'_{0,\lambda}}^j (\omega_{\text{CY}} - \omega'_{0,\lambda}) \right|_{\omega'_{0,\lambda}} = O(r^{-\beta-j}).$$

Consequently, ω_{CY} is of generalized ALG type with cone angle $\theta = \frac{2\pi}{k}$.

Example 1. *Suppose that $\dim_{\mathbb{C}} A = 2$, so that $\dim_{\mathbb{C}} X = 3$. Every stabilizer group over $t = 0$ is a finite cyclic subgroup of $\text{SL}(3, \mathbb{C})$, and hence the corresponding local quotient singularity is Gorenstein. By Roan's theorem [17], every three-dimensional Gorenstein quotient singularity admits a crepant resolution. In the present cyclic setting, the quotient singularities are toric, and the local crepant resolutions may be chosen projective over the quotient. Thus the abstract existence assumption in Theorem 4.1 is satisfied for the examples below.*

The following Table 1 summarize the cyclic actions associated with Ueno's classification of singular fibers of normally polarized abelian surfaces [21, 22], together with the corresponding cone angles. The fourth column demonstrates the candidates of isotrivial fibers (for each type, we only list one possibility of isotrivial fiber). The fifth column describes the associated Ueno fiber types of singular fibers at $t = 0$ and $t = \infty$ respectively. Case 8 is an additional model that does not appear explicitly in Ueno's classification.

In Cases 1–4, the resolution is locally the product of the resolution of a surface quotient with a smooth elliptic curve. These examples may therefore also be understood by taking the product of

Case	k	Generator of action on (t, w_1, w_2)	Isotrivial Fiber	Ueno types at $0, \infty$	Cone angle
1	2	$(t, w_1, w_2) \mapsto (-t, -w_1, w_2)$	$E_{\text{arb}} \times E_{\text{arb}}$	II①(a) at both ends	π
2	3	$(t, w_1, w_2) \mapsto (\zeta_3 t, \zeta_3^2 w_1, w_2)$	$E_{\text{hex}} \times E_{\text{arb}}$	III②(b) and III②(a)	$2\pi/3$
3	4	$(t, w_1, w_2) \mapsto (it, -iw_1, w_2)$	$E_{\text{sq}} \times E_{\text{arb}}$	III⑤(a) and III⑤(b)	$\pi/2$
4	6	$(t, w_1, w_2) \mapsto (\zeta_6 t, \zeta_6^5 w_1, w_2)$	$E_{\text{hex}} \times E_{\text{arb}}$	III④(a) and III④(b)	$\pi/3$
5	3	$(t, w_1, w_2) \mapsto (\zeta_3 t, \zeta_3 w_1, \zeta_3 w_2)$	$E_{\text{hex}} \times E_{\text{hex}}$	II③(a) and II③(b)	$2\pi/3$
6	4	$(t, w_1, w_2) \mapsto (it, iw_1, -w_2)$	$E_{\text{sq}} \times E_{\text{arb}}$	III⑥(a) and III⑥(b)	$\pi/2$
7	6	$(t, w_1, w_2) \mapsto (\zeta_6 t, \zeta_3 w_1, -w_2)$	$E_{\text{hex}} \times E_{\text{arb}}$	III③(a) and III③(b)	$\pi/3$
8	6	$(t, w_1, w_2) \mapsto (\zeta_6 t, \zeta_6 w_1, \zeta_3^2 w_2)$	$E_{\text{hex}} \times E_{\text{hex}}$	not listed by Ueno	$\pi/3$
9	12	$(t, w_1, w_2) \mapsto (\zeta_{12} t, \zeta_6 w_1, -iw_2)$	$E_{\text{hex}} \times E_{\text{sq}}$	IV⑦ at both ends	$\pi/6$
10	12	$(t, w_1, w_2) \mapsto (\zeta_{12} t, iw_1, \zeta_3^2 w_2)$	$E_{\text{sq}} \times E_{\text{hex}}$	IV⑥ at both ends	$\pi/6$
11	5	$(t, w_1, w_2) \mapsto (\zeta_5 t, \zeta_5 w_1, \zeta_5^3 w_2)$	$\mathbb{Z}_5$ -abelian surface	IV⑥ at both ends	$2\pi/5$
12	6	$(t, w_1, w_2) \mapsto (\zeta_6 t, \zeta_3^2 w_2, \zeta_3^2 w_1)$	$E_{\text{hex}} \times E_{\text{hex}}$	IV② at both ends	$\pi/3$
13	6	$(t, w_1, w_2) \mapsto (\zeta_6 t, \zeta_6 w_2, \zeta_6 w_1)$	$E_{\text{hex}} \times E_{\text{hex}}$	IV② at both ends	$\pi/3$

Table 1. The isotrivial threefold models. k denotes the order of the action. $E_{\text{hex}} = \mathbb{C}/(\mathbb{Z} + \zeta_3 \mathbb{Z})$, $E_{\text{sq}} = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$ and E_{arb} denotes an arbitrary elliptic curve. In Case 11, the abelian surface is $A = \mathbb{C}^2/\Lambda$, where $\Lambda = \langle (1, 1), (\zeta_5, \zeta_5^2), (\zeta_5^2, \zeta_5^4), (\zeta_5^3, \zeta_5) \rangle$.

Hein's metrics with a flat elliptic-curve metric. Cases 5–13 contain genuinely three-dimensional quotient singularities or nonisolated singular strata, but the stratified QALE construction in the proof of Theorem 4.1 treats all of them uniformly.

Proof of Theorem 4.1. We divide the argument into three steps: the geometric construction, the extension of the quotient metric across the exceptional locus, and the solution of the complex Monge–Ampère equation.

Step1: The geometric construction. Let $q : \mathbb{P}^1 \times A \rightarrow Y$ be the quotient map. On $\mathbb{C}_t \times A$, consider the holomorphic n -form $\Theta = dt \wedge \Omega_A$. Since $\gamma^* dt = \zeta_k dt$, $\tau^* \Omega_A = \zeta_k^{-1} \Omega_A$, the form Θ is G -invariant. It therefore descends to a rational n -form Θ_Y on Y .

The morphism π is crepant over the image of $t = 0$. Hence $\pi^* \Theta_Y$ extends across the exceptional locus without acquiring either zeros or poles. Since π is an isomorphism near the fiber over $t = \infty$, the divisor of $\pi^* \Theta_Y$ can be computed directly on the covering space near $t = \infty$.

Put $s = t^{-1}$. Then $\Theta = -\frac{1}{s^2} ds \wedge \Omega_A$. Thus the descended form has a pole of order two along the reduced fiber over infinity. On the quotient base, a local coordinate at infinity is $z = s^k$. Consequently, the scheme-theoretic fiber is

$$F = \text{div}(z) = kF_{\text{red}}.$$

Since τ has order k , a generic point of A has trivial stabilizer. Hence F_{red} is the irreducible quotient $A/\langle \tau \rangle$. We obtain

$$\text{div}(\pi^* \Theta_Y) = -2F_{\text{red}},$$

and therefore

$$K_X \sim_{\mathbb{Q}} -2F_{\text{red}} = -\frac{2}{k}F.$$

This proves the canonical divisor formula without requiring local coordinates on the quotient.

The action of G is free away from the fibers over 0 and ∞ . Since all quotient singularities over 0 are resolved, X is smooth on M . The singularities remaining along F are finite quotient

singularities and are therefore klt. Moreover, Y is projective because it is a finite quotient of the projective variety $\mathbb{P}^1 \times A$, and X is projective because π is projective.

If C is a curve contained in a fiber of f , then $F \cdot C = 0$. It follows from (4.1) that $K_X \cdot C = 0$. Thus K_X is relatively nef and f is relatively minimal.

We next construct the model metric. Choose a translation-invariant flat Kähler form ω_A on A representing a polarization. Replacing ω_A by $\frac{1}{k} \sum_{j=0}^{k-1} (\tau^j)^* \omega_A$ if necessary, we may assume that ω_A is τ -invariant. For an appropriate constant $c_\lambda > 0$, the product metric $\widehat{\omega}_\lambda = \frac{ic_\lambda}{2} dt \wedge d\bar{t} + \omega_A$ satisfies $\widehat{\omega}_\lambda^n = n! \lambda i^{n^2} \Theta \wedge \bar{\Theta}$. The form $\widehat{\omega}_\lambda$ is G -invariant and therefore descends to the smooth locus of Y .

On each smooth fiber, the descended metric is the unique flat metric in the prescribed polarization class and with the prescribed volume. Its horizontal distribution is the flat horizontal distribution inherited from the product $\mathbb{C}_t \times A$. By the uniqueness in the semi-flat construction, the descended product metric is exactly the semi-flat metric. In particular, no calculation involving period matrices or local coordinates is required to identify the semi-flat ansatz.

Step2: The extension of the quotient metric. It remains to extend this quotient metric across the exceptional locus over $t = 0$. We use the stratification of quotient singularities developed by Joyce in [11]. Let $y \in Y$ lie over $t = 0$, and choose a point $\widehat{y} \in \{0\} \times A$ mapping to y . Let $H = \text{Stab}_G(\widehat{y})$. By holomorphic linearization of finite group actions, an analytic neighborhood of y is modeled on a neighborhood of the origin in $\mathbb{C}^n/H$. For every $h \in H$, the determinant of its action on the tangent space equals its action on the holomorphic volume form. Since $dt \wedge \Omega_A$ is H -invariant, we have $H \subset \text{SL}(n, \mathbb{C})$.

We recall the local stratification. Let $\mathcal{L}_H = \{\text{Fix}(K) : K \subseteq H\}$ be the finite collection of fixed subspaces in $\mathbb{C}^n$. Choose an indexing set I and write $\mathcal{L}_H = \{V_i : i \in I\}$. The indexing set is partially ordered by $i \preceq j \Leftrightarrow V_i \subseteq V_j$. Let $W_i = V_i^\perp$, so that $\mathbb{C}^n = V_i \oplus W_i$, and define $A_i = \{h \in H : h|_{V_i} = \text{id}_{V_i}\}$. At a generic point of the stratum determined by V_i , the quotient is locally modeled on $V_i \times (W_i/A_i)$.

By [11, Proposition 3.2], every crepant resolution of $\mathbb{C}^n/H$ is a local product resolution. Consequently, the restriction of π to the chosen quotient chart determines a crepant resolution $\pi_i : Y_i \rightarrow W_i/A_i$ of the transverse quotient singularity. Moreover, each Y_i is itself a local product crepant resolution.

Let T_i be a sufficiently small neighborhood of the union of the strata properly contained in V_i , and let U_i be its inverse image in $V_i \times Y_i$. The local product structure is expressed by the commutative diagram

$$\begin{array}{ccc} (V_i \times Y_i) \setminus U_i & \xrightarrow{\psi_i} & X \\ \text{id}_{V_i} \times \pi_i \downarrow & & \downarrow \pi \\ (V_i \times (W_i/A_i)) \setminus T_i & \xrightarrow{\phi_i} & Y, \end{array} \quad (4.2)$$

where ψ_i and ϕ_i are local analytic isomorphisms onto their images. The square is Cartesian over the indicated open sets.

We now construct the extension of the quotient metric by induction over the partially ordered set of strata. For a deepest stratum, the transverse quotient W_i/A_i has an isolated singularity. On its crepant resolution Y_i , choose a Kähler metric which agrees exactly with the Euclidean quotient metric outside a compact set. This is the initial step of the QALE construction.

Suppose inductively that compatible Kähler metrics have been chosen on the transverse resolutions associated with all strata properly contained in V_i . Since Y_i is itself a local product crepant resolution, [11, Theorem 4.7] gives a QALE Kähler metric g_i on Y_i which has the following properties:

- (1) g_i agrees exactly with the Euclidean quotient metric away from a fixed neighborhood of the exceptional locus;
- (2) near every lower-dimensional stratum, g_i is exactly the product of the Euclidean metric along the stratum and the already chosen metric on the corresponding transverse resolution;
- (3) g_i may be chosen in the Kähler class induced by a projective Kähler form on X .

Let $B_i = N_H(V_i)/A_i$ be the residual finite group acting on the transverse resolution. Averaging the relevant Kähler potentials over B_i , we may assume that g_i is B_i -invariant. The product metric $h_{V_i} \oplus g_i$ on $V_i \times Y_i$, where h_{V_i} is the Euclidean metric on V_i , therefore descends through the local product chart (4.2).

The compatibility on overlaps follows from the inductive construction. Indeed, if $i \preceq j$, then the resolution Y_i is itself locally a product involving Y_j , and the metric g_i is exactly the corresponding product metric outside a prescribed transition region. Thus the models associated with V_i and V_j coincide wherever both descriptions apply.

Equivalently, let φ_i denote the local Kähler potentials which modify the pullback of the quotient metric on the transverse resolutions. They satisfy the equivariance and compatibility conditions in [11, Proposition 4.4]. Joyce's gluing theorem [11, Theorem 4.5 and Proposition 4.6] produces a global potential φ such that $\omega_{0,\lambda} = \pi^*\omega_{q,\lambda} + i\partial\bar{\partial}\varphi$ is positive, where $\omega_{q,\lambda}$ denotes the quotient of $\widehat{\omega}_\lambda$. The cutoff functions in this construction depend only on the distances to the deeper strata and may be chosen invariant under all residual finite-group actions.

Since the singular locus over $t = 0$ is compact, only finitely many quotient charts are required. The local constructions agree on overlaps by the preceding compatibility, and they agree exactly with $\pi^*\omega_{q,\lambda}$ near the boundaries of the chosen neighborhoods. Hence they glue to a global Kähler metric $\omega_{0,\lambda}$ on M satisfying

$$\omega_{0,\lambda} = \omega_{q,\lambda} = \omega_{\text{sf},\lambda}$$

outside a compact neighborhood of the exceptional locus over $t = 0$.

Near infinity, the metric is therefore the quotient of a product of the Euclidean plane and the flat torus A . A fundamental sector in the Euclidean factor has opening angle $\frac{2\pi}{k}$, and its two boundary faces are identified by the automorphism τ . Thus $\omega_{0,\lambda}$ is exactly a generalized ALG model of cone angle $2\pi/k$ outside a compact set.

Step3: The solution of the complex Monge–Ampère equation.

We finally solve the complex Monge–Ampère equation. Using the compactly supported adjustment described in Section 6, we modify $\omega_{0,\lambda}$ inside a compact set to obtain a complete Kähler metric $\omega'_{0,\lambda}$ which remains unchanged near infinity and satisfies the required integral constraint. Define its Ricci potential $\mathfrak{f}$ by

$$e^{\mathfrak{f}}(\omega'_{0,\lambda})^n = n!\lambda i^{n^2}\Omega \wedge \overline{\Omega}.$$

Since $\omega'_{0,\lambda}$ agrees with the normalized flat model near infinity, the function $\mathfrak{f}$ is compactly supported. The adjustment is chosen so that

$$\int_M (e^{\mathfrak{f}} - 1) (\omega'_{0,\lambda})^n = 0.$$

The exact flat structure at infinity gives a quasi-atlas of uniformly controlled holomorphic coordinate charts. The metric has quadratic volume growth, a uniform Ricci lower bound, and uniformly connected annuli. It therefore satisfies the SOB(2) condition. The Tian–Yau–Hein theorem [8, Proposition 4.1], gives a smooth function u such that $\omega_{\text{CY}} = \omega'_{0,\lambda} + i\partial\bar{\partial}u$ is complete and satisfies

$$\omega_{\text{CY}}^n = e^{\mathfrak{f}}(\omega'_{0,\lambda})^n = n!\lambda i^{n^2}\Omega \wedge \overline{\Omega}.$$

Because the background metric is exactly an isotrivial generalized ALG model outside a compact set and $\mathfrak{f}$ is compactly supported, the Schwartz-decay result [23, Proposition 1.6] applies. Therefore,

for every $\beta > 0$ and every $j \in \mathbb{N}_0$,

$$\left| \nabla_{\omega'_{0,\lambda}}^j i\partial\bar{\partial}u \right|_{\omega'_{0,\lambda}} = O(r^{-\beta-j}).$$

This proves the third assertion, and completes the proof. $\square$

5. CONSTRUCTION OF GENERALIZED ALG, ALG*, ALH, AND ALH* TYPE ANSATZ

In this section, we introduce the non-isotrivial geometric models underlying Theorem 5.1. The central innovation is a carefully arranged fiber-product construction: a generic fiber product does not possess the geometry required for our metric construction, whereas matching suitable Kodaira singular fibers, while keeping all remaining singular fibers disjoint, produces the desired models. We recall the relevant local descriptions, construct projective relatively minimal compactifications with at worst klt singularities, compute their canonical divisors, and identify the resulting complete ends as being of generalized ALG, ALH, ALG*, or ALH* type.

5.1. Elaboration on the main theorem. We now state the main result for the non-isotrivial fiber-product models.

Theorem 5.1 (non-isotrivial constructions). *Let $\pi_i : X_i \rightarrow \mathbb{P}^1$, $i = 1, 2$, be two rational elliptic fibrations. For each i , fix a singular fiber F_i of π_i . Assume that F_1 and F_2 lie over the same point $p \in \mathbb{P}^1$ and that the remaining singular values of π_1 and π_2 are disjoint. Consider the fiber product $f : X = X_1 \times_{\mathbb{P}^1} X_2 \rightarrow \mathbb{P}^1$. Then the singular locus of X is contained in the fiber $F = F_1 \times F_2$. Set $M = X \setminus F$. We have the following.*

(1) *Suppose that the monodromies of both F_1 and F_2 are finite.*

(1a) *Let n_i be the order of the local monodromy of F_i and set $k = \text{lcm}(n_1, n_2)$. After the minimal simultaneous base change $z = s^k$, the pullback over the punctured disk has the form*

$$\mathcal{U}' = (\Delta_s^* \times \mathbb{C}_{w_1, w_2}^2) / \Lambda(s),$$

where $\Lambda(s)$ is the lattice generated by the four period sections. The original fibration is $\mathcal{U}' / \langle \mathcal{A} \rangle$, where the distinguished generator $s \mapsto \zeta_k s$ acts by

$$\mathcal{A} : (s, w_1, w_2) \mapsto (\zeta_k s, \zeta_k^\alpha h_1(s) w_1, \zeta_k^\beta h_2(s) w_2).$$

Here $h_i(s)$ is a holomorphic unit determined by the corresponding elliptic fibration and normalized by $h_i(0) = 1$, while $1 \leq \alpha, \beta < k$ are the deck weights specified in Table 4 below.

(1b) *The fibration $f|_M : M \rightarrow \mathbb{P}^1 \setminus \{p\}$ admits a normal projective compactification $\tilde{f} : \tilde{X} \rightarrow \mathbb{P}^1$ such that $\tilde{X}$ has at worst klt singularities and $\tilde{f}$ is relatively minimal. Moreover,*

$$K_{\tilde{X}} \sim_{\mathbb{Q}} \frac{k - \alpha - \beta - 1}{k} [\tilde{F}],$$

where $\tilde{F}$ denotes any fiber of $\tilde{f}$.

(1c) *If $\alpha + \beta > k$, then M admits a complete Calabi–Yau metric of generalized ALG type with cone angle $\theta = \frac{2(\alpha + \beta - k)\pi}{k}$.*

(1d) *If $\alpha + \beta = k$, then M admits a complete Calabi–Yau metric of generalized ALH type.*

(2) *Suppose that F_i is of type $I_{b_i}^*$, where $b_i > 0$ for $i = 1, 2$,*

(2a) *The fibration $f|_M : M \rightarrow \mathbb{P}^1 \setminus \{p\}$ admits a normal projective compactification $\tilde{f} : \tilde{X} \rightarrow \mathbb{P}^1$ such that $\tilde{X}$ has at worst klt singularities and $\tilde{f}$ is relatively minimal. Moreover,*

$$K_{\tilde{X}} \sim_{\mathbb{Q}} -\frac{1}{2} [\tilde{F}],$$

where $\tilde{F}$ denotes any fiber of $\tilde{f}$.

- (2b) *There exists a complete Calabi–Yau metric ω_{CY} on M of generalized ALH_{b_1, b_2}^* type. In particular, $\text{Vol}(B(x_0, s)) \sim s^{\frac{3}{2}}$ for $s \gg 1$, and its unique tangent cone at infinity is $\mathbb{R}_+$.*
- (3) *Suppose that F_1 is of type I_b^* , where $b > 0$, and that F_2 is of type II^* , III^* , or IV^* .*
- (3a) *The fibration $f|_M : M \rightarrow \mathbb{P}^1 \setminus \{p\}$ admits a normal projective compactification $\tilde{f} : \tilde{X} \rightarrow \mathbb{P}^1$ such that $\tilde{X}$ has at worst klt singularities and $\tilde{f}$ is relatively minimal. Moreover,*

$$K_{\tilde{X}} \sim_{\mathbb{Q}} \begin{cases} -\frac{1}{2}[\tilde{F}], & \text{if } F_2 \text{ is of type } \text{II}^*, \\ -\frac{1}{2}[\tilde{F}], & \text{if } F_2 \text{ is of type } \text{III}^*, \\ -\frac{1}{3}[\tilde{F}], & \text{if } F_2 \text{ is of type } \text{IV}^*, \end{cases}$$

where $\tilde{F}$ denotes any fiber of $\tilde{f}$.

- (3b) *There exists a complete Calabi–Yau metric ω_{CY} on M of generalized $\text{ALG}_{b, T}^*$ type, where T denotes the Kodaira type of F_2 . In particular, $\text{Vol}(B(x_0, s)) \sim s^2$ for $s \gg 1$. Its unique tangent cone at infinity is a flat metric cone of angle $2\pi/3$, $\pi/2$, or $\pi/3$ when F_2 is of type II^* , III^* , or IV^* , respectively.*

Remark 5.2. *One may try to use the adjunction formula to conclude that the fiber product of two rational elliptic surfaces is already a Calabi–Yau variety. In the cases considered here, however, the fiber product is nonnormal, so it does not carry a canonical Weil divisor in the usual sense. The failure of normality is caused by the nonreduced components of the distinguished singular fibers, for example those of type II^* , III^* , IV^* , or I_b^* .*

The rest of this section is devoted to the construction of the ansatz at the end of the models described in Theorem 5.1 and the analysis of the behavior of the ansatz. See Section 6 for how to extend the ansatz to global ansatz and perturb it to genuine Calabi–Yau metrics.

5.2. Review of local models on elliptic fibrations. Since our construction of the non-isotrivial ansatz relies heavily on the fiber product structure of two elliptic fibrations, we review the local models of singular fibers in elliptic fibrations. We follow the arguments in [8, 13].

Consider an elliptic fibration $\pi : U \rightarrow \Delta$ with only one singular fiber, lying over 0. Then $U|_{\Delta^*}$ is isomorphic to a quotient

$$(\Delta_z^* \times \mathbb{C}_v) / (\mathbb{Z}\tau_1(z) + \mathbb{Z}\tau_2(z)),$$

where the $\tau_i(z)$ are multivalued functions and (z, v) are local coordinates. A holomorphic volume form Ω on U can be written over $U|_{\Delta^*}$ as

$$\Omega = g(z) dz \wedge dv.$$

The coordinates (z, v) do not extend across the singular fiber. If (x, y) are local holomorphic coordinates near a smooth point of that fiber, then

$$dz \wedge dv = (\pi^* z)^N h(x, y) dx \wedge dy,$$

where h is nowhere vanishing. Thus N is the vanishing order of $dz \wedge dv$ along the scheme-theoretic fiber. Since the semi-flat ansatz is expressed in the coordinates (z, v) , determining N is essential.

Case 1: Finite monodromy. We begin by recalling the local model around a singular fiber with finite monodromy in an elliptic fibration. Let $\pi : U \rightarrow \Delta_z$ be an elliptic fibration, such that $z = 0$ is the only singular fiber whose monodromy is finite.

We treat only a singular fiber of type IV ; other cases can be treated similarly. According to Kodaira, the j -invariant has a zero at $z = 0$. Thus $\mathcal{J}(0) = 0$; let $m := \text{mult}_0 \mathcal{J} \in \mathbb{N}$. From Kodaira

[13], we know that $m \equiv 2 \pmod{3}$. When τ approaches ζ_3 , we can perform a coordinate change on Δ , so that

$$\tau(z) = \zeta_3 \frac{1 - \zeta_3 z^{\frac{m}{3}}}{1 - z^{\frac{m}{3}}}$$

Traversing a counterclockwise loop around $z = 0$ once induces a transformation on τ given by

$$\tau \mapsto A\tau = \frac{d\tau + b}{c\tau + a}$$

where $A = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}$.

The pullback U' of $U|_{\Delta^*}$ under $z = s^3$ can be written as $(\Delta_s^* \times \mathbb{C}_w) / (\mathbb{Z} + \mathbb{Z}\tau(s^3))$ and thus extends over $s = 0$ with central fiber $\mathbb{C} / (\mathbb{Z} + \mathbb{Z}\zeta_3)$. The deck action of the monodromy on U' is generated by

$$\mathcal{A}(s, w) = \left(\zeta_3 s, \zeta_3 \frac{1 - s^m}{1 - \zeta_3^2 s^m} w \right)$$

Note that

$$\Phi : \Delta^* \times \mathbb{C} \rightarrow \Delta^* \times \mathbb{C}, (s, w) \mapsto (s^3, (1 - s^m) s^2 w)$$

factors through the monodromy action A , thus inducing an isomorphism of elliptic fibrations

$$\Phi : U|_{\Delta^*} = U' / \sim_{\mathcal{A}} \longrightarrow (\Delta^* \times \mathbb{C}) / (\mathbb{Z}\tau_1 + \mathbb{Z}\tau_2),$$

where

$$\begin{aligned} \tau_1(z) &= \text{pr}_2 \Phi \left(z^{\frac{1}{3}}, 1 \right) = (1 - z^{\frac{m}{3}}) z^{\frac{2}{3}}; \\ \tau_2(z) &= \text{pr}_2 \Phi \left(z^{\frac{1}{3}}, \zeta_3 \right) = \zeta_3 (1 - \zeta_3 z^{\frac{m}{3}}) z^{\frac{2}{3}}. \end{aligned}$$

Moreover, $(z, v) = (s^3, (1 - s^m) s^2 w)$ gives coordinates on $U|_{\Delta^*}$. By analyzing the canonical resolution as in [8], one finds that the vanishing order of $dz \wedge dv$ along the scheme-theoretic singular fiber is $N = 1$.

For the reader's convenience, we list all the cases in the following table:

Recall that $m \equiv 1 \pmod{3}$ for type II and IV*; $m \equiv 2 \pmod{3}$ for type II* and IV; $m \equiv 1 \pmod{2}$ for type III and III*.

Case 2: I_b singular fiber. Let $\pi : U \rightarrow \Delta$ be an elliptic fibration with I_1 singular fiber at 0. According to Kodaira, the map

$$\Psi : (\Delta^* \times \mathbb{C}) / (\mathbb{Z}\tau_1 + \mathbb{Z}\tau_2) \rightarrow \Delta_z \times \mathbb{P}_{[X:Y:W]}^2, (z, v) \mapsto \left(z, \left[-\frac{1}{12} - \frac{1}{4\pi^2} \wp_z(v) : \frac{i}{8\pi^3} \wp'_z(v) : 1 \right] \right)$$

is an isomorphism between

$$(\Delta^* \times \mathbb{C}) / (\mathbb{Z}\tau_1 + \mathbb{Z}\tau_2) \cong \{(z, [X : Y : W]) | Y^2 W = 4X^3 + X^2 W - g_2(z) X W^2 - g_3(z) W^3\} \subset \Delta^* \times \mathbb{P}^2$$

where $\tau_1 = 1$, $\tau_2 = \frac{1}{2\pi i} \log z$, $g_2(z) = 20 \sum_{n=1}^{\infty} (1 - z^n)^{-1} n^3 z^n$, $g_3(z) = \frac{1}{3} \sum_{n=1}^{\infty} (1 - z^n)^{-1} (7n^5 + 5n^3) z^n$ are regular on Δ with $g_2(0) = g_3(0) = 0$ and $\wp_z(v) = \frac{1}{v^2} + \sum_{\lambda \in \Lambda(z) \setminus \{0\}} \left(\frac{1}{(v-\lambda)^2} - \frac{1}{\lambda^2} \right)$ is the Weierstrass

$\wp$ -function associated with the lattice $\Lambda(z)$ generated by $\tau_1(z), \tau_2(z)$. $\text{Im } \Psi$ in $\Delta \times \mathbb{P}^2$ is a smooth elliptic surface $\bar{U}$ with central fiber the nodal cubic, $y^2 = 4x^3 + x^2$. This $\bar{U}$ represents the Kodaira canonical form for a singular fiber of type I_1 .

$\pi^*(0)$	Matrix A	Deck action $\mathcal{A}$	Coordinate (z, v)	Generators τ_1, τ_2
I_0^*	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	$(s, w) \mapsto (-s, -w)$	(s^2, sw)	$z^{\frac{1}{2}}, z^{\frac{1}{2}}\tau(z)$
II	$\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$	$(s, w) \mapsto \left(\zeta_6 s, \zeta_6 \frac{1-s^{2m}}{1-\zeta_3 s^{2m}} w\right)$	$(s^6, (1-s^{2m})s^5 w)$	$(1-z^{\frac{m}{3}})z^{\frac{5}{6}}, \zeta_3(1-\zeta_3 z^{\frac{m}{3}})z^{\frac{5}{6}}$
II*	$\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$	$(s, w) \mapsto \left(\zeta_6 s, \zeta_6^5 \frac{1-s^{2m}}{1-\zeta_3 s^{2m}} w\right)$	$(s^6, (1-s^{2m})sw)$	$(1-z^{\frac{m}{3}})z^{\frac{1}{6}}, \zeta_3(1-\zeta_3 z^{\frac{m}{3}})z^{\frac{1}{6}}$
III	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	$(s, w) \mapsto \left(is, i \frac{1-s^{2m}}{1+s^{2m}} w\right)$	$(s^4, (1-s^{2m})s^3 w)$	$(1-z^{\frac{m}{2}})z^{\frac{3}{4}}, i(1+z^{\frac{m}{2}})z^{\frac{3}{4}}$
III*	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$(s, w) \mapsto \left(is, -i \frac{1-s^{2m}}{1+s^{2m}} w\right)$	$(s^4, (1-s^{2m})sw)$	$(1-z^{\frac{m}{2}})z^{\frac{1}{4}}, i(1+z^{\frac{m}{2}})z^{\frac{1}{4}}$
IV	$\begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}$	$(s, w) \mapsto \left(\zeta_3 s, \zeta_3 \frac{1-s^m}{1-\zeta_3 s^m} w\right)$	$(s^3, (1-s^m)s^2 w)$	$(1-z^{\frac{m}{3}})z^{\frac{2}{3}}, \zeta_3(1-\zeta_3 z^{\frac{m}{3}})z^{\frac{2}{3}}$
IV*	$\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$	$(s, w) \mapsto \left(\zeta_3 s, \zeta_3^2 \frac{1-s^m}{1-\zeta_3 s^m} w\right)$	$(s^3, (1-s^m)sw)$	$(1-z^{\frac{m}{3}})z^{\frac{1}{3}}, \zeta_3(1-\zeta_3 z^{\frac{m}{3}})z^{\frac{1}{3}}$

Table 2. Local quotient data for finite-monodromy Kodaira fibers.

Recall that $\frac{dx}{y}$ is the generating holomorphic 1-form on an elliptic curve $E = \{[X : Y : W] \in \mathbb{P}^2 \mid Y^2 W = 4X^3 + X^2 W - g_2(z)XW^2 - g_3(z)W^3\}$, where $x = \frac{X}{W}$ and $y = \frac{Y}{W}$. From

$$\Psi^* \left(dz \wedge \frac{dx}{y} \right) = dz \wedge \frac{-\frac{1}{4\pi^2} d\wp_z(v)}{\frac{i}{8\pi^3} \wp'_z(v)} = 2\pi i dz \wedge dv,$$

we see that $dz \wedge dv$ extends without vanishing along the singular fiber; hence $N = 0$.

An I_b singular fiber can be represented over Δ^* by $\tau_1 = 1$ and $\tau_2 = \frac{b}{2\pi i} \log z$. It admits a fiber-preserving, unramified b -fold covering onto an I_1 degeneration over Δ^* . Thus $N = 0$ in this case.

Case 3: I_b^* singular fiber. Let $\pi : U \rightarrow \Delta$ be an elliptic fibration with I_b^* singular fiber at 0. The pullback of $U|_{\Delta^*}$ under $z = u^2$ extends as $U' \rightarrow \Delta_u$ with an I_{2b} central fiber. $\mathbb{Z}_2$ acts naturally on U' , extending the action by deck transformations on U' , in such a way that the quotient elliptic surface has four ordinary double points along its central fiber. Resolving these singularities results in the Kodaira canonical form $\bar{U}$ for I_b^* . We can assume that $U'|_{\Delta^*} = (\Delta_u^* \times \mathbb{C}_w) / (\mathbb{Z}\tau_1 + \mathbb{Z}\tau_2)$ with $\tau_1 = 1$ and $\tau_2 = \frac{b}{\pi i} \log u$. Since $du \wedge dw$ is invariant under the $\mathbb{Z}_2$ -action, it induces a holomorphic volume form on the crepant resolution $\bar{U}$.

Now we can change coordinates

$$\Delta_u^* \times \mathbb{C}_w \rightarrow \Delta_z^* \times \mathbb{C}_v, (u, w) \mapsto (z, v) = (u^2, uw).$$

In the coordinates (z, v) , we have

$$\tau_1(z) = z^{\frac{1}{2}}, \quad \tau_2(z) = \frac{b}{2\pi i} z^{\frac{1}{2}} \log z, \quad dz \wedge dv = 2u^2 du \wedge dw.$$

Therefore, the vanishing order of $dz \wedge dv$ along the scheme-theoretic singular fiber is $N = 1$.

5.3. Ansatz on fiber product.

Proof of part of Theorem 5.1: compactifications and asymptotic models. We first establish the compactification statements and identify the corresponding semi-flat models according to Kodaira

types. The global gluing and Monge–Ampère perturbation are carried out in Proposition 6.1 and Corollary 6.2.

Let $p \in \mathbb{P}^1$ be the common image of F_1 and F_2 , and choose $q \in \mathbb{P}^1 \setminus \{p\}$ over which both elliptic fibrations are smooth. For each positive integer n used below, let $\rho_n : C_n \rightarrow \mathbb{P}^1$ be the cyclic cover of degree n totally ramified over p and q , with deck group $G_n \cong \mathbb{Z}_n$. Thus every local base change $z = s^n$ appearing below is induced by a global algebraic cover.

We repeatedly use the following elementary cyclic-quotient observation. Let a cyclic group G_k act on a smooth total space over Δ_s , let the base coordinate on the quotient satisfy $z = s^k$, and suppose that the action is generically free along every component of the reduced central fiber $\widehat{D}$. If D is its image in the quotient, then the scheme-theoretic quotient fiber is kD . Indeed, at a general point $x \in \widehat{D}$ one may choose a neighborhood disjoint from all its nontrivial translates. The quotient map is therefore biholomorphic on that neighborhood, where s is a local coordinate and $z = s^k$. The same argument shows that, for every invariant rational top form Θ ,

$$\text{ord}_D(\Theta) = \text{ord}_{\widehat{D}}(\varpi^* \Theta), \quad (5.1)$$

where ϖ denotes the quotient map.

Situation 1: Both F_i are of $\text{I}_{b_i}^*$ type, corresponding to (2) in Theorem 5.1.

(2a) Let U_i be a neighborhood of the singular fiber F_i . Let Ω_i be the holomorphic volume form on $X_i \setminus F_i$. From the discussion in Subsection 5.2, there exist appropriate coordinates (z_i, v_i) on U_i such that $\Omega_i = \frac{k_i(z_i)}{z_i^2} dz_i \wedge dv_i$, where $i = 1, 2$ and $k_i(0) \neq 0$. By the Poincaré residue formula, we obtain a holomorphic volume form Ω on $X \setminus F$ such that, in the fiber-product coordinates,

$$\Omega = \frac{k(z)}{z^2} dz \wedge dv_1 \wedge dv_2.$$

Consider the double cover $\rho_2 : C_2 \rightarrow \mathbb{P}^1$, and let $\widehat{X}_i \rightarrow C_2$ be the relatively minimal regular model of the normalization of $X_i \times_{\mathbb{P}^1} C_2$. The fiber of $\widehat{X}_i$ over $\widehat{p}$ is of type I_{2b_i} . The fiber product $\widehat{X} = \widehat{X}_1 \times_{C_2} \widehat{X}_2$ is a normal projective threefold. Its only singularities over $\widehat{p}$ are the $4b_1b_2$ ordinary double points locally given by $xy - uv = 0$. In particular, $\widehat{X}$ has terminal, and hence klt, singularities. The deck involution acts algebraically on $\widehat{X}$, and $\widetilde{X} = \widehat{X}/G_2$ is a normal projective compactification of M . Since a finite quotient of a klt variety is klt, $\widetilde{X}$ has at worst klt singularities.

On the double cover, the coordinates are related by

$$(z, v_1, v_2) = (u^2, uw_1, uw_2).$$

Therefore,

$$\Omega = \frac{k(z)}{z^2} dz \wedge dv_1 \wedge dv_2 = \frac{2k(u^2)}{u} du \wedge dw_1 \wedge dw_2.$$

Since $z = u^2$, the scheme-theoretic fiber satisfies $\widetilde{F} = 2\widetilde{F}_{\text{red}}$. The preceding expression has a simple pole along $\widetilde{F}_{\text{red}}$. It follows that

$$K_{\widetilde{X}} \sim_{\mathbb{Q}} -\frac{1}{2}[\widetilde{F}].$$

Since $\widetilde{F} = \widetilde{f}^*(p)$, we have $K_{\widetilde{X}} \cdot C = 0$ for every curve C contracted by $\widetilde{f}$. Hence $K_{\widetilde{X}}$ is relatively nef. Together with the klt property established above, this shows that $\widetilde{f} : \widetilde{X} \rightarrow \mathbb{P}^1$ is relatively minimal.

(2b) The periods in the coordinates (z, v_1, v_2) are

$$\tau_1(z) = z^{\frac{1}{2}}, \quad \tau_2(z) = \frac{b_1}{2\pi i} z^{\frac{1}{2}} \log z, \quad \tau_3(z) = z^{\frac{1}{2}}, \quad \tau_4(z) = \frac{b_2}{2\pi i} z^{\frac{1}{2}} \log z.$$

Substituting these data into (2.1) yields

$$\begin{aligned}\omega_{\text{sf},\varepsilon} = & \frac{ib_1b_2|k|^2|\log|z||^2}{\pi^2\varepsilon^2|z|^2}dz \wedge d\bar{z} \\ & + \frac{i\pi\varepsilon}{|z||\log|z||} \sum_{j=1}^2 \frac{1}{b_j} \left\{ dv_j - \left(\frac{v_j}{2z} - \frac{i\operatorname{Im}\left(z^{-\frac{1}{2}}v_j\right)}{z^{\frac{1}{2}}|\log|z||} \right) dz \right\} \\ & \wedge \left\{ d\bar{v}_j - \left(\frac{\bar{v}_j}{2\bar{z}} + \frac{i\operatorname{Im}\left(z^{-\frac{1}{2}}v_j\right)}{\bar{z}^{\frac{1}{2}}|\log|z||} \right) d\bar{z} \right\}.\end{aligned}$$

To analyze the asymptotic behavior, we rewrite the ansatz using the coordinates (u, w_1, w_2) :

$$\begin{aligned}\omega_{\text{sf},\varepsilon} = & \frac{4i|k|^2|\log|u||^2}{\pi^2\varepsilon^2|u|^2}du \wedge d\bar{u} \\ & + \frac{i\pi\varepsilon}{2|\log|u||} \sum_{j=1}^2 \frac{1}{b_j} \left(dw_j + \frac{i\operatorname{Im}(w_j)}{u|\log|u||} du \right) \wedge \left(d\bar{w}_j - \frac{i\operatorname{Im}(w_j)}{\bar{u}|\log|u||} d\bar{u} \right).\end{aligned}$$

Let $\Delta^* = \{z \in \mathbb{C} : 0 < |z| < \frac{1}{2}\}$, $U = f^{-1}(\Delta^*)$, and $z_x = f(x)$ for $x \in U$. Fix $x_0 \in U$ and set $r_x = \operatorname{dist}_{g_{\text{sf},\varepsilon}}(x_0, x)$. Let

$$g_B = \frac{2b_1b_2|k(0)|^2|\log|z||^2}{\pi^2\varepsilon^2|z|^2}|dz|^2$$

be the Riemannian metric associated with the leading horizontal part of the semi-flat form.

The diameter of $f^{-1}(z)$ is approximately $C|\log|u||^{\frac{1}{2}} = C'|\log|z||^{\frac{1}{2}}$. The g_B -length of $\{z' \in \Delta^* : |z'| = |z|\}$ is approximately $2\pi|\log|z||$. For fixed z_0 and any z lying on the radial segment through z_0 , the radial distance from z_0 to z is approximately $\left| \int_{|z_0|}^{|z|} \frac{\log r}{r} dr \right| \sim \frac{1}{2}|\log|z||^2$ as $z \rightarrow 0$. Thus, for any fixed $x_0 \in U$ and any $x \in U$ sufficiently close to the singular fiber, we have

$$r_x \sim \operatorname{dist}_{g_B}(z_x, z_{x_0}) \sim |\log|z_x||^2.$$

The map $f : U \rightarrow \Delta^*$ is a Riemannian submersion whose fiber volume is ε^2 . Hence, for any $B \subset \Delta^*$,

$$\operatorname{Vol}(f^{-1}(B), g_{\text{sf},\varepsilon}) = \varepsilon^2 \operatorname{Vol}(B, g_B).$$

Comparing geodesic balls with preimages of radial base regions gives

$$\operatorname{Vol}_{g_{\text{sf},\varepsilon}}(B(x_0, r_x)) \sim C \int_0^{2\pi} d\theta \int_{|z_x|}^{|z_{x_0}|} \frac{(\log r)^2}{r^2} r dr \sim C |\log|z_x||^3 \sim r_x^{\frac{3}{2}},$$

Thus, the volume growth of geodesic balls is of order $r^{\frac{3}{2}}$.

Next, we prove $|B(x, \frac{1}{2}r_x)| \geq \frac{1}{C}r_x^{\frac{3}{2}}$ if $r_x \gg 1$. Recall that the diameter of $f^{-1}(z)$ and the g_B -length of $\{z' \in \Delta : |z'| = |z|\}$ are of order $|\log|z||^{\frac{1}{2}}$ and $|\log|z||$, respectively. Thus

$$B\left(x, \frac{1}{2}r_x\right) \supset \{y \in U : |z_x| < |z_y| < |z_x|^{1-\alpha}\}$$

for all $x \in U$, $r_x \gg 1$, and sufficiently small $\alpha > 0$. By the coarea formula,

$$\left| B\left(x, \frac{1}{2}r_x\right) \right| \geq C \int_{|z_x|}^{|z_x|^{1-\alpha}} \frac{|\log r|^2}{r} dr \sim |\log|z_x||^3 \sim r_x^{\frac{3}{2}}.$$

Together with the Ricci-flatness of $\omega_{\text{sf},\varepsilon}$ and the annular connectivity evident from the model geometry, the preceding distance and volume estimates verify all the conditions of SOB $(\frac{3}{2})$. Put

$L = |\log |z||$. Since $r \sim L^2$, the angular direction has length $O(L) = O(r^{\frac{1}{2}})$, while the fiber diameter is $O(L^{\frac{1}{2}}) = O(r^{\frac{1}{4}})$. Consequently, every connected component of a sufficiently large annular level set is contained in a ball of radius $O(r^{\frac{1}{2}})$. Thus the end satisfies CYL $(\frac{3}{2}, \frac{1}{2})$. The two collapsing torus directions have length of order $L^{-\frac{1}{2}}$, and therefore $|B(x, 1)| \sim L^{-1} \sim r^{-\frac{1}{2}}$. Hence the polynomial unit-ball condition in [8, Proposition 4.8 (ib)] fails for $0 < \kappa < \frac{1}{2}$, whereas $\exp(\kappa r^{\frac{1}{2}})|B(x, 1)| \rightarrow \infty$ for every $\kappa > 0$. The appropriate decay statement for this end is therefore [8, Proposition 4.8 (ia)], using the CYL $(\frac{3}{2}, \frac{1}{2})$ condition.

To determine the norm of the Riemann curvature tensor, we only need to calculate the norm of the Chern curvature. Direct computation shows that

$$|\text{Rm}(x)|^2 \sim \frac{(\pi\varepsilon)^4 |z_x|^2}{4|k(0)|^4 |\log |z_x||^4} \sim e^{-cr_x^{\frac{1}{2}}} r_x^{-2}.$$

Together with the corresponding bounds for all derivatives in the lifted coordinate charts, this gives the required $C^{k,\alpha}$ quasi-atlas. The displayed curvature estimate implies $|\text{Rm}| = O(r^{-2})$. Since the semi-flat metric is Ricci-flat, [8, Lemma 4.7], applied with $\lambda = 1$, shows that it is HMG(1, $k+1$, γ) for every $k \in \mathbb{N}_0$ and every $\gamma \in (0, 1)$. Thus the CYL condition supplies the initial oscillation estimate in [8, Proposition 4.8 (ia)], while the HMG condition supplies the scale-invariant local regularity estimates used to obtain derivative decay. The preceding cross-sectional diameter estimates also imply the at-most-linear diameter condition in [8, Proposition 4.8 (ii)].

Now we compute the asymptotic cone of M . Since the diameter of $f^{-1}(z_x)$ is only of order $r_x^{\frac{1}{4}}$, it is enough to compute the tangent cone of g_B . Define

$$\Phi_\lambda : [0, \infty) \times S^1 \rightarrow \Delta^*, (s, \theta) \mapsto \frac{1}{2} \exp \left(-\lambda^{-\frac{1}{2}} s^{\frac{1}{2}} + i\theta \right) = z.$$

A direct computation shows that

$$\begin{aligned} \Phi_\lambda^*(\lambda^2 g_B) &= \frac{2b_1 b_2 |k(z)|^2 \lambda^2}{\pi^2 \varepsilon^2} \left| \log \frac{1}{2} - \lambda^{-\frac{1}{2}} s^{\frac{1}{2}} \right|^2 \left(\frac{1}{4} \lambda^{-1} s^{-1} ds \otimes ds + d\theta \otimes d\theta \right) \\ &\xrightarrow{\lambda \rightarrow 0} \frac{b_1 b_2 |k(0)|^2}{2\pi^2 \varepsilon^2} ds \otimes ds. \end{aligned}$$

Hence the tangent cone is $\mathbb{R}^+$.

Situation 2: F_1 is of I_b^* type and F_2 is of II^* , III^* , or IV^* type, corresponding to (3) in Theorem 5.1.

Let $T \in \{\text{II}^*, \text{III}^*, \text{IV}^*\}$ be the type of F_2 , and retain the notation n_T, l_T from (5.6). Set

$$k = \text{lcm}(2, n_T), \quad d_T = \frac{kl_T}{n_T}, \quad \sigma_T = \frac{1}{2} - \frac{l_T}{n_T}. \quad (5.2)$$

The complete numerical data are recorded in Table 3.

F_1	$F_2 = T$	k	(α, β)	$(k/2, d_T)$	$\text{ord}_D \Omega$	Canonical divisor and asymptotic model
I_b^*	II^*	6	(3, 5)	(3, 1)	-3	$-\frac{1}{2}\tilde{F}$; $\text{ALG}_{b, \text{II}^*}^*$, angle $\frac{2\pi}{3}$
I_b^*	III^*	4	(2, 3)	(2, 1)	-2	$-\frac{1}{2}\tilde{F}$; $\text{ALG}_{b, \text{III}^*}^*$, angle $\frac{\pi}{2}$
I_b^*	IV^*	6	(3, 4)	(3, 2)	-2	$-\frac{1}{3}\tilde{F}$; $\text{ALG}_{b, \text{IV}^*}^*$, angle $\frac{\pi}{3}$

Table 3. The mixed $I_b^* \times T$ cases. Here $(z, v_1, v_2) = (s^k, s^{k/2} w_1, s^{d_T} w_2')$ up to holomorphic units, and $D = \tilde{F}_{\text{red}}$.

Take the degree- k cyclic cover $C_k \rightarrow \mathbb{P}^1$ introduced above. On the relatively minimal regular models of the normalized base changes, the first central fiber is of type I_{kb} and the second fibration

has good reduction. At all other points, the two discriminant sets are disjoint. Hence their fiber product $\widehat{X}$ is a smooth projective threefold. Its quotient

$$\widetilde{X} := \widehat{X}/G_k$$

is a normal projective compactification of M with only quotient singularities and therefore has at worst klt singularities.

The diagonal deck action is generically free along every component of the central fiber. For $T = \text{II}^*$ or III^* , the action of the generator on the good-reduction elliptic factor already has order k , so every nonidentity element has a proper fixed locus on that factor. For $T = \text{IV}^*$, its action on the good-reduction factor has order 3; the only additional element to consider is the involution g^3 . On the normalization of the I_{6b} polygon, this is the stable-reduction involution $x \mapsto x^{-1}$, which also sends the component indexed by j to the component indexed by $-j$. It therefore fixes no component pointwise. Thus the hypothesis of the cyclic-quotient observation is satisfied in all three cases.

After absorbing a holomorphic unit into w'_2 , the quotient coordinates are

$$(z, v_1, v_2) = (s^k, s^{k/2}w_1, s^{d_T}w'_2).$$

If $D = \widetilde{F}_{\text{red}}$, then $\widetilde{F} = kD$. Pulling back the residue form (5.8) gives

$$\varpi^*\Omega = k\kappa(s^k)U_T(s)s^{d_T-k/2-1}ds \wedge dw_1 \wedge dw'_2, \quad U_T(0) \neq 0. \quad (5.3)$$

Using (5.1), we obtain

$$K_{\widetilde{X}} \sim_{\mathbb{Q}} \frac{d_T - k/2 - 1}{k} \widetilde{F} = \begin{cases} -\frac{1}{2}\widetilde{F}, & T = \text{II}^*, \text{III}^*, \\ -\frac{1}{3}\widetilde{F}, & T = \text{IV}^*. \end{cases}$$

This divisor is numerically trivial over the base. Together with the klt property, this proves that $\widetilde{f}$ is relatively minimal and establishes (3a) in all three cases.

We next identify the semi-flat geometry uniformly. The period areas satisfy

$$\text{Im}(\overline{\tau}_1\tau_2) = \frac{b}{2\pi}|z||\log|z||, \quad \text{Im}(\overline{\tau}_3\tau_4) = c_T|z|^{2l_T/n_T}(1 + O(|z|^\nu))$$

for some $c_T, \nu > 0$, with analogous estimates for all derivatives. Consequently, the horizontal metric has the form

$$g_B = C_T|\log|z|||z|^{-2-2\sigma_T}|dz|^2(1 + O(|z|^\nu)). \quad (5.4)$$

Writing $\rho = |z|$, the distance to a fixed compact set satisfies

$$r \sim C'_T|\log\rho|^{1/2}\rho^{-\sigma_T}. \quad (5.5)$$

The ratio of the length of the circle $\rho = \text{const}$ to r therefore converges to $2\pi\sigma_T$. The fiber diameter is $O(|\log\rho|^{1/2}) = o(r)$, while its volume is constant. It follows that the volume growth is quadratic and that the unique tangent cone at infinity is the flat cone of angle

$$2\pi\sigma_T = \begin{cases} \frac{2\pi}{3}, & T = \text{II}^*, \\ \frac{\pi}{2}, & T = \text{III}^*, \\ \frac{\pi}{3}, & T = \text{IV}^*. \end{cases}$$

Together with the period description, this is precisely the generalized $\text{ALG}_{b,T}^*$ model of Definition 3.6.

For completeness, the same estimates verify all analytic hypotheses used in Section 6. The annular distance and volume estimates following from (5.4) give SOB(2) and the at-most-linear diameter condition. Exactly one torus direction has length comparable to $|\log\rho|^{-1/2}$, so

$$|B(x, 1)| \sim |\log\rho|^{-1/2}.$$

By (5.5), $r^\delta |B(x, 1)| \rightarrow \infty$ for every $\delta > 0$. Direct differentiation of the uniform semi-flat formula in the lifted charts gives $|\text{Rm}| = O(r^{-2})$, together with the corresponding bounds for all derivatives. Since the model is Ricci-flat, [8, Lemma 4.7] implies $\text{HMG}(1, j+1, \gamma)$ for every $j \in \mathbb{N}_0$ and $\gamma \in (0, 1)$. Proposition 6.1 and Corollary 6.2 then give the complete Calabi–Yau metrics asserted in (3b).

We have therefore verified the analytic hypotheses needed below. The generalized ALG and ALG^* models satisfy the relevant SOB condition and the unit-ball hypothesis in [8, Proposition 4.8 (ib)]. The generalized ALH model satisfies both SOB(1) and $\text{CYL}(1, 0)$, while the generalized ALH^* model satisfies $\text{CYL}(\frac{3}{2}, \frac{1}{2})$ and falls under [8, Proposition 4.8 (ia)]. All these model metrics are $\text{HMG}(1, k+1, \gamma)$ for every $k \in \mathbb{N}_0$ and every $\gamma \in (0, 1)$, and all satisfy the at-most-linear diameter condition. These properties are unchanged by compactly supported modifications and therefore also hold for any global glued metric that agrees with the corresponding model metric near infinity.

Situation 3: Both F_1 and F_2 are of finite monodromy, corresponding to (1) in Theorem 5.1.

For a finite-monodromy Kodaira type T , let n_T be the order of its monodromy. After multiplying the fiber coordinate by a holomorphic unit, the stable-reduction data can be written as

$$(s, w') \mapsto (\zeta_{n_T} s, \zeta_{n_T}^{a_T} w'), \quad (z, v) = (s^{n_T}, s^{l_T} w'), \quad l_T = n_T - a_T. \quad (5.6)$$

The local models in Subsection 5.2 give

$$(n_T, a_T, l_T) = (2, 1, 1), (6, 1, 5), (6, 5, 1), (4, 1, 3), (4, 3, 1), (3, 1, 2), (3, 2, 1)$$

for

$$T = \text{I}_0^*, \text{II}, \text{II}^*, \text{III}, \text{III}^*, \text{IV}, \text{IV}^*,$$

respectively. Let T_i be the type of F_i and set

$$k = \text{lcm}(n_{T_1}, n_{T_2}), \quad \alpha = \frac{k}{n_{T_1}} a_{T_1}, \quad \beta = \frac{k}{n_{T_2}} a_{T_2}.$$

Thus k is the degree of the minimal simultaneous base change giving good reduction for both factors. On the common cover, after absorbing the holomorphic units into w'_1 and w'_2 , the generator acts by

$$(s, w'_1, w'_2) \mapsto (\zeta_k s, \zeta_k^\alpha w'_1, \zeta_k^\beta w'_2),$$

and the quotient coordinates satisfy

$$(z, v_1, v_2) = (s^k, s^{k-\alpha} w'_1, s^{k-\beta} w'_2). \quad (5.7)$$

Before this harmless change of fiber coordinates, the two holomorphic units are precisely the factors $h_i(s)$ in Theorem 5.1. This proves (1a) for every pair of finite-monodromy fibers.

Let $\widehat{X}_i \rightarrow C_k$ be the relatively minimal regular model of the normalization of $X_i \times_{\mathbb{P}^1} C_k$. Both fibrations have good reduction over the point $\widehat{p}$ above p . At every other point of C_k , at least one of the two morphisms $\widehat{X}_i \rightarrow C_k$ is a submersion: the remaining discriminant sets are disjoint, and both original fibrations are smooth over the second branch point q . It follows that

$$\widehat{X} := \widehat{X}_1 \times_{C_k} \widehat{X}_2$$

is a smooth projective threefold. The deck action extends algebraically, and

$$\widetilde{X} := \widehat{X}/G_k$$

is a normal projective compactification of M with only quotient singularities. In particular, $\widetilde{X}$ has at worst klt singularities.

The central fiber of $\widehat{X}$ is a product of two smooth elliptic curves. The induced automorphisms of the two factors have orders n_{T_1} and n_{T_2} , so their diagonal action has exact order k . The fixed locus of every nonidentity element is proper; hence the action is free at a general point of the central fiber. Let D denote the reduced fiber of $\widetilde{X}$ over p and $\widetilde{F}$ denote a general fiber. The cyclic-quotient observation above gives $\widetilde{F} \sim kD$.

The Poincaré residue construction gives a nowhere-vanishing holomorphic volume form on M of the form

$$\Omega = \frac{\kappa(z)}{z^2} dz \wedge dv_1 \wedge dv_2, \quad \kappa(0) \neq 0. \quad (5.8)$$

Pulling this form back by the quotient map and using (5.7), we obtain

$$\varpi^* \Omega = k\kappa(s^k)U(s)s^{k-\alpha-\beta-1} ds \wedge dw'_1 \wedge dw'_2,$$

where $U(0) \neq 0$. Indeed, every derivative of a unit or of a power of s occurring in dv_i is multiplied by ds and therefore disappears after wedging with ds . By (5.1) and $\tilde{F} \sim kD$,

$$K_{\tilde{X}} \sim_{\mathbb{Q}} (k - \alpha - \beta - 1)D = \frac{k - \alpha - \beta - 1}{k} \tilde{F}.$$

Since $\tilde{F} = \tilde{f}^*(p)$, this divisor has zero intersection with every curve contracted by $\tilde{f}$. Thus $K_{\tilde{X}}$ is relatively nef, and the klt fibration $\tilde{f}: \tilde{X} \rightarrow \mathbb{P}^1$ is relatively minimal. This proves (1b) uniformly.

For reference, Table 4 records every pair. In its fifth column, c is the coefficient in $K_{\tilde{X}} \sim_{\mathbb{Q}} c\tilde{F}$.

It remains to identify the complete ends in Table 4. Using the deck actions and the period generators listed in Table 2, and then passing to the minimal simultaneous cover $z = s^k$, we may write

$$(\tau_1, \tau_2) = z^{(k-\alpha)/k}(u_1, u_2), \quad (\tau_3, \tau_4) = z^{(k-\beta)/k}(u_3, u_4),$$

after choosing branches on the universal cover of Δ^* . Here the u_i are holomorphic functions of the appropriate finite-cover coordinate, and $(u_1(0), u_2(0))$ and $(u_3(0), u_4(0))$ generate nondegenerate limiting lattices. It follows that the two period areas have the uniform leading behavior

$$\operatorname{Im}(\bar{\tau}_1 \tau_2) = c_1 |z|^{2(k-\alpha)/k} (1 + \varepsilon_1(z)), \quad \operatorname{Im}(\bar{\tau}_3 \tau_4) = c_2 |z|^{2(k-\beta)/k} (1 + \varepsilon_2(z)), \quad (5.9)$$

where $c_1, c_2 > 0$ and $\varepsilon_i(z) = O(|z|^\nu)$ for some $\nu > 0$. Moreover, for every $p, q \in \mathbb{Z}_{\geq 0}$,

$$(z\partial_z)^p (\bar{z}\partial_{\bar{z}})^q \varepsilon_i(z) = O(|z|^\nu).$$

These estimates follow directly from the holomorphic dependence on the finite-cover coordinate s , since $z\partial_z = k^{-1}s\partial_s$. In particular, the normalized period areas converge to their leading terms with all logarithmic derivatives, which gives the required C^∞ control of the semi-flat asymptotic model. Put $\sigma = \frac{\alpha+\beta-k}{k}$. It follows directly from the semi-flat formula in Corollary 2.5 that its horizontal metric satisfies

$$g_B = C|z|^{-2-2\sigma}|dz|^2(1 + O(|z|^\nu)) \quad (5.10)$$

for some $C > 0$.

If $\sigma > 0$, set $r = C^{1/2}\sigma^{-1}|z|^{-\sigma}$. The leading term in (5.10) becomes $dr^2 + \sigma^2 r^2 d\theta^2$. After also introducing the normalized fiber coordinates

$$\beta_1 = z^{-(k-\alpha)/k}v_1, \quad \beta_2 = z^{-(k-\beta)/k}v_2,$$

the semi-flat metric converges with all derivatives to a flat product metric on a sector times a flat four-torus. Analytic continuation around $z = 0$ identifies the boundary rays by the limiting torus isometry. The end is therefore generalized ALG with cone angle

$$2\pi\sigma = \frac{2(\alpha + \beta - k)\pi}{k}.$$

If $\sigma = 0$, the coordinate $-\log z$ gives a flat cylindrical leading term, and the finite monodromy produces the compact flat mapping-torus cross-section allowed in Definition 3.1. The convergence is exponential with all derivatives, so the end is generalized ALH. If $\sigma < 0$, the radial integral is finite, explaining the incomplete entries in Table 4.

These estimates also verify the analytic hypotheses used in Section 6. In the ALG cases the metric satisfies SOB(2), has uniformly noncollapsed unit balls, and has at-most-linear cross-sectional diameter. In the ALH cases it satisfies SOB(1) and CYL(1,0). In both cases the all-derivative

F_1	F_2	k	(α, β)	c	Semi-flat behavior
I_0^*	I_0^*	2	(1, 1)	$-\frac{1}{2}$	ALH
I_0^*	II	6	(3, 1)	$\frac{1}{6}$	incomplete
I_0^*	II^*	6	(3, 5)	$-\frac{1}{2}$	ALG, angle $\frac{2\pi}{3}$
I_0^*	III	4	(2, 1)	0	incomplete
I_0^*	III^*	4	(2, 3)	$-\frac{1}{2}$	ALG, angle $\frac{\pi}{2}$
I_0^*	IV	6	(3, 2)	0	incomplete
I_0^*	IV^*	6	(3, 4)	$-\frac{1}{3}$	ALG, angle $\frac{\pi}{3}$
II	II	6	(1, 1)	$\frac{1}{2}$	incomplete
II	II^*	6	(1, 5)	$-\frac{1}{6}$	ALH
II	III	12	(2, 3)	$\frac{1}{2}$	incomplete
II	III^*	12	(2, 9)	0	incomplete
II	IV	6	(1, 2)	$\frac{1}{3}$	incomplete
II	IV^*	6	(1, 4)	0	incomplete
II^*	II^*	6	(5, 5)	$-\frac{5}{6}$	ALG, angle $\frac{4\pi}{3}$
II^*	III	12	(10, 3)	$-\frac{1}{6}$	ALG, angle $\frac{\pi}{6}$
II^*	III^*	12	(10, 9)	$-\frac{2}{3}$	ALG, angle $\frac{7\pi}{6}$
II^*	IV	6	(5, 2)	$-\frac{1}{3}$	ALG, angle $\frac{\pi}{3}$
II^*	IV^*	6	(5, 4)	$-\frac{2}{3}$	ALG, angle π
III	III	4	(1, 1)	$\frac{1}{4}$	incomplete
III	III^*	4	(1, 3)	$-\frac{1}{4}$	ALH
III	IV	12	(3, 4)	$\frac{1}{3}$	incomplete
III	IV^*	12	(3, 8)	0	incomplete
III^*	III^*	4	(3, 3)	$-\frac{3}{4}$	ALG, angle π
III^*	IV	12	(9, 4)	$-\frac{1}{6}$	ALG, angle $\frac{\pi}{6}$
III^*	IV^*	12	(9, 8)	$-\frac{1}{2}$	ALG, angle $\frac{5\pi}{6}$
IV	IV	3	(1, 1)	0	incomplete
IV	IV^*	3	(1, 2)	$-\frac{1}{3}$	ALH
IV^*	IV^*	3	(2, 2)	$-\frac{2}{3}$	ALG, angle $\frac{2\pi}{3}$

Table 4. The finite-monodromy pairs. The two fibers may be interchanged. Here *incomplete* means that the distinguished puncture is at finite distance for the semi-flat metric, so (1c) and (1d) make no assertion.

convergence implies $|\text{Rm}| = O(r^{-2})$; hence [8, Lemma 4.7] gives $\text{HMG}(1, j+1, \gamma)$ for every $j \in \mathbb{N}_0$ and $\gamma \in (0, 1)$. Proposition 6.1 and Corollary 6.2 then yield the complete Calabi–Yau metrics asserted in (1c) and (1d).

Remark 5.3. Since $X \subset X_1 \times X_2$ is a hypersurface, the adjunction formula for the dualizing sheaf gives

$$\omega_X \cong (K_{X_1 \times X_2} \otimes \mathcal{O}_{X_1 \times X_2}(X))|_X.$$

The residue construction above trivializes this invertible sheaf. When X is nonnormal, however, one cannot interpret this statement as the triviality of a canonical Weil divisor. The Poincaré residue Ω is defined on the smooth manifold M and gives a global trivialization of K_M .

This distinction is important for the construction of complete Calabi–Yau metrics. If every irreducible component of both F_1 and F_2 is reduced, as happens for fibers of type I_b , II , III , and IV , then the fiber product X is normal and K_X is indeed trivial. Moreover, in the cases $I_n \times I_m$, $I_n \times II$, $I_n \times III$, $I_n \times IV$, $II \times II$, $II \times III$, $III \times III$, and $IV \times IV$, the results of [12, 18] provide a crepant resolution of X . The form Ω then lifts to a global holomorphic volume form on the resolution. Consequently, the present method does not produce a complete Calabi–Yau metric on the complement of the distinguished fiber in these cases.

6. PERTURBATION OF THE SEMI-FLAT ANSATZ

In this section, we extend the semi-flat ansatz constructed in the previous section to a global Kähler metric and then apply the Tian–Yau–Hein package to perturb it to a complete Calabi–Yau metric. There are two issues. The first is a cohomological $\partial\bar{\partial}$ -problem on the end. The second is to preserve positivity during the gluing while simultaneously satisfying the integral normalization required by the Tian–Yau–Hein package.

Let $f : X \rightarrow \mathbb{P}^1$ be one of the abelian fibrations constructed in Section 5, let $F = f^{-1}(p)$ be the distinguished singular fiber, and set $M = X \setminus F$. Fix a sufficiently small disk $\Delta \subset \mathbb{P}^1$ centered at p , over whose punctured part $\Delta^* = \Delta \setminus \{p\}$ the fibration is smooth. Write $X|_{\Delta^*} = (\Delta_z^* \times \mathbb{C}^2) / \Lambda(z)$. For $\varepsilon > 0$ and $\alpha > 0$, let $\omega_{\text{sf},\varepsilon}(\alpha)$ be the semi-flat metric constructed in the previous section, normalized by $\omega_{\text{sf},\varepsilon}(\alpha)^3 = 6i\alpha \Omega \wedge \bar{\Omega}$.

We first state the global gluing result.

Proposition 6.1. *For every $\alpha > 0$, there exist a holomorphic section $\tilde{\sigma}$ of f over Δ^* , concentric disks $\Delta' \subset \Delta'' \subset \Delta''' \subset \Delta$ and a complete Kähler metric $\omega_0(\alpha)$ on M with the following properties:*

(1) *Over Δ'^* ,*

$$\omega_0(\alpha) = T^* \omega_{\text{sf},\varepsilon}(\alpha),$$

where T is the fiberwise translation determined by $\tilde{\sigma}$ relative to the initially chosen section.

(2) *There exists a Kähler form ω on M such that*

$$\omega_0(\alpha) = \omega$$

over $\mathbb{P}^1 \setminus \Delta'''$.

(3) *The integral normalization*

$$\int_M (\omega_0(\alpha)^3 - 6i\alpha \Omega \wedge \bar{\Omega}) = 0 \tag{6.1}$$

is satisfied.

Proof. We divide the proof into two steps.

Step 1: Cohomological matching and the $\partial\bar{\partial}$ -problem.

Let $\hat{F}$ be a smooth fiber over Δ^* , and let

$$h_q : H_q(\hat{F}, \mathbb{R}) \longrightarrow H_q(\hat{F}, \mathbb{R})$$

denote the monodromy action. Since Δ^* deformation retracts onto a circle, $X|_{\Delta^*}$ is homotopy equivalent to the mapping torus of the monodromy. Set

$$H_2(\hat{F}, \mathbb{R})_{h_2} := \frac{H_2(\hat{F}, \mathbb{R})}{(h_2 - \text{id})H_2(\hat{F}, \mathbb{R})}, \quad H_1(\hat{F}, \mathbb{R})^{h_1} := \ker(h_1 - \text{id}).$$

The Wang exact sequence [6, Proposition 6.4.8, p. 212] then yields the short exact sequence

$$0 \longrightarrow H_2(\widehat{F}, \mathbb{R})_{h_2} \xrightarrow{\bar{\iota}_*} H_2(X|_{\Delta^*}, \mathbb{R}) \xrightarrow{\partial_W} H_1(\widehat{F}, \mathbb{R})^{h_1} \longrightarrow 0, \quad (6.2)$$

where $\bar{\iota}_*$ is induced by the inclusion of a smooth fiber and ∂_W is the connecting homomorphism in the Wang sequence. The classes in the last term are represented by mapping-torus 2-cycles obtained by transporting monodromy-invariant 1-cycles around the puncture. These are the cycles sometimes called *bad cycles* in the literature.

In our situation,

$$M = X_1^\circ \times_{\mathbb{P}^1 \setminus \{p\}} X_2^\circ, \quad X_i^\circ = X_i \setminus F_i,$$

where $X_i \rightarrow \mathbb{P}^1$ are rational elliptic surfaces. By the construction in [8, Chapter 5], one may choose Kähler forms ω_i on X_i° whose restrictions represent the prescribed semi-flat fiber classes and whose periods over all mapping-torus 2-cycles vanish. Let $p_i : M \rightarrow X_i^\circ$ be the natural projections and define $\omega = p_1^* \omega_1 + p_2^* \omega_2$. This is a Kähler form on M .

For a smooth fiber $\widehat{F} = \widehat{F}_1 \times \widehat{F}_2$, the Künneth decomposition gives

$$H_2(\widehat{F}, \mathbb{R}) \cong H_2(\widehat{F}_1, \mathbb{R}) \oplus \left(H_1(\widehat{F}_1, \mathbb{R}) \otimes H_1(\widehat{F}_2, \mathbb{R}) \right) \oplus H_2(\widehat{F}_2, \mathbb{R}).$$

Both ω and the semi-flat form are induced by the product polarization. Their periods agree on the first and third summands and vanish on the mixed summand.

Let

$$h_{1,i} : H_1(\widehat{F}_i, \mathbb{R}) \longrightarrow H_1(\widehat{F}_i, \mathbb{R}), \quad i = 1, 2,$$

denote the monodromy actions of the two elliptic factors. Under the natural identification

$$H_1(\widehat{F}, \mathbb{R}) \cong H_1(\widehat{F}_1, \mathbb{R}) \oplus H_1(\widehat{F}_2, \mathbb{R}),$$

one has $h_1 = h_{1,1} \oplus h_{1,2}$ and hence

$$\ker(h_1 - \text{id}) = \ker(h_{1,1} - \text{id}) \oplus \ker(h_{1,2} - \text{id}).$$

The periods of both forms over the corresponding mapping-torus cycles vanish. It follows from (6.2) that

$$[\omega_{\text{sf}, \varepsilon}(1)] = [\omega] \quad \text{in} \quad H^2(X|_{\Delta^*}, \mathbb{R}). \quad (6.3)$$

Consequently, there is a real 1-form ζ such that

$$\omega_{\text{sf}, \varepsilon}(1) - \omega = d\zeta.$$

Set $\xi = \zeta^{0,1}$. Since the difference is of type $(1, 1)$, $\bar{\partial}\xi = 0$. Because Δ^* is Stein, the Leray spectral sequence and Cartan's Theorem B give

$$H^{0,1}(X|_{\Delta^*}) \cong H^1(X|_{\Delta^*}, \mathcal{O}_{X|_{\Delta^*}}) \cong H^0(\Delta^*, R^1 f_* \mathcal{O}_{X|_{\Delta^*}}). \quad (6.4)$$

Thus $[\xi]$ is determined by its fiberwise Dolbeault classes. Using the polarization, this section may be represented by a holomorphic section of the relative Lie algebra. Exponentiating it gives a holomorphic section $\tilde{\sigma}$ of the torus fibration. The standard translation calculation for semi-flat forms then shows that, after translating by $\tilde{\sigma}$, the class in (6.4) vanishes. The remaining term pulled back from Δ^* has a smooth real potential. Hence there exists $u_1 \in C^\infty(X|_{\Delta^*}, \mathbb{R})$ such that

$$T^* \omega_{\text{sf}, \varepsilon}(1) - \omega = i\partial\bar{\partial}u_1. \quad (6.5)$$

The dependence of the semi-flat metric on α occurs entirely in its horizontal component. The explicit formula from the preceding section therefore gives a smooth function $u_{B, \alpha}$ on Δ^* such that

$$T^* \omega_{\text{sf}, \varepsilon}(\alpha) - T^* \omega_{\text{sf}, \varepsilon}(1) = i\partial\bar{\partial}f^* u_{B, \alpha}.$$

After setting $u_\alpha = u_1 + f^* u_{B, \alpha}$, we obtain

$$T^* \omega_{\text{sf}, \varepsilon}(\alpha) - \omega = i\partial\bar{\partial}u_\alpha \quad (6.6)$$

for every $\alpha > 0$.

Step 2: Positivity and integral normalization.

Choose concentric disks $\Delta_1 \subset \Delta_2 \subset \Delta$ and a cutoff function χ pulled back from the base such that $\chi \equiv 1$ on Δ_1 , $\chi \equiv 0$ outside Δ_2 . Define the closed real $(1, 1)$ -form

$$\tilde{\omega}_\alpha = \omega + i\partial\bar{\partial}(\chi u_\alpha). \quad (6.7)$$

It agrees with $T^*\omega_{\text{sf},\varepsilon}(\alpha)$ over Δ_1^* and with ω outside Δ_2 .

Although $\tilde{\omega}_\alpha$ need not be positive on the transition region, its restriction to every fiber is positive. Indeed, because χ is pulled back from the base,

$$\tilde{\omega}_\alpha|_{X_z} = (1 - \chi(z)) \omega|_{X_z} + \chi(z) T^*\omega_{\text{sf},\varepsilon}(\alpha)|_{X_z}.$$

This is a convex combination of two positive fiber metrics.

The transition region is compact. Since the vertical block of $\tilde{\omega}_\alpha$ is uniformly positive there, the Schur complement criterion shows that one can choose a nonnegative $(1, 1)$ -form κ_α on the base, supported in $\Delta_2 \setminus \overline{\Delta_1}$, such that $\hat{\omega}_\alpha = \tilde{\omega}_\alpha + f^*\kappa_\alpha$ is Kähler. Thus positivity can always be restored by adding a base form supported in the compact gluing region; no lower bound on α is involved.

It remains to impose the integral normalization. Define

$$D_\alpha = \int_M (\hat{\omega}_\alpha^3 - 6i\alpha \Omega \wedge \bar{\Omega}).$$

This integral is finite because the two volume forms agree near infinity.

Let $V_\varepsilon = \int_{X_z} \hat{\omega}_\alpha^2|_{X_z}$. The number $V_\varepsilon > 0$ is independent of z and of all base corrections. If ϑ is any compactly supported real $(1, 1)$ -form on the base, then $(f^*\vartheta)^2 = 0$, and hence

$$\int_M (\hat{\omega}_\alpha + f^*\vartheta)^3 - \int_M \hat{\omega}_\alpha^3 = 3 \int_M \hat{\omega}_\alpha^2 \wedge f^*\vartheta = 3V_\varepsilon \int_{\mathbb{P}^1 \setminus \{p\}} \vartheta.$$

If $D_\alpha \leq 0$, choose a nonnegative base form ϑ_α supported in the compact transition region such that

$$\int_{\mathbb{P}^1} \vartheta_\alpha = -\frac{D_\alpha}{3V_\varepsilon}.$$

Then $\hat{\omega}_\alpha + f^*\vartheta_\alpha$ remains Kähler and satisfies (6.1).

Suppose instead that $D_\alpha > 0$. On the region where $\hat{\omega}_\alpha = T^*\omega_{\text{sf},\varepsilon}(\alpha)$, write

$$T^*\omega_{\text{sf},\varepsilon}(\alpha) = \omega_{\text{B},\alpha} + \omega_{\text{F},\varepsilon},$$

where $\omega_{\text{B},\alpha}$ is the horizontal base component. The explicit model formula gives $\int_{\Delta^*} \omega_{\text{B},\alpha} = +\infty$. Consequently, one may choose $0 < \delta_1 < \delta_2$ and a smooth nonnegative base form ϑ_α satisfying

$$\text{supp}(\vartheta_\alpha) \subset \left\{ \frac{\delta_1}{2} < |z| < 2\delta_2 \right\}, \quad 0 \leq \vartheta_\alpha \leq \frac{1}{2}\omega_{\text{B},\alpha},$$

and

$$\int_{\mathbb{P}^1} \vartheta_\alpha = \frac{D_\alpha}{3V_\varepsilon}.$$

Indeed, one first chooses an annulus on which one half of the model base form has integral larger than $D_\alpha/(3V_\varepsilon)$ and then multiplies by an appropriate cutoff function. It follows that $\hat{\omega}_\alpha - f^*\vartheta_\alpha$ remains Kähler and satisfies (6.1).

In either case, denote the resulting metric by $\omega_0(\alpha)$. All corrections are compactly supported. After shrinking the innermost disk and enlarging the outermost disk if necessary, $\omega_0(\alpha)$ satisfies all the conclusions of the theorem. Since it agrees with the complete semi-flat model near infinity, it is complete. $\square$

Corollary 6.2. *For every $\alpha > 0$, there exists a smooth function u on M such that $\omega_{\text{CY}} = \omega_0(\alpha) + i\partial\bar{\partial}u$ is a complete Ricci-flat Kähler metric satisfying*

$$\omega_{\text{CY}}^3 = 6i\alpha \Omega \wedge \bar{\Omega}.$$

Moreover, if

$$\omega_{\text{mod}} = T^* \omega_{\text{sf},\varepsilon}(\alpha)$$

denotes the model metric on the end, then there exists $\delta > 0$ such that, for every $j \geq 0$,

$$|\nabla_{\omega_{\text{mod}}}^j (\omega_{\text{CY}} - \omega_{\text{mod}})|_{\omega_{\text{mod}}} = O(r^{-2-j-\delta}).$$

In particular, ω_{CY} has the same asymptotic model as $\omega_{\text{sf},\varepsilon}(\alpha)$.

Proof. Define the Ricci potential of the glued metric by

$$f_\alpha = \log \frac{6i\alpha \Omega \wedge \bar{\Omega}}{\omega_0(\alpha)^3}.$$

Since $\omega_0(\alpha) = \omega_{\text{mod}}$ near infinity and $\omega_{\text{mod}}^3 = 6i\alpha \Omega \wedge \bar{\Omega}$, the function f_α is compactly supported. Moreover, Theorem 6.1 gives

$$\int_M (e^{f_\alpha} - 1) \omega_0(\alpha)^3 = \int_M (6i\alpha \Omega \wedge \bar{\Omega} - \omega_0(\alpha)^3) = 0.$$

All the geometric hypotheses of the Tian-Yau-Hein's package [8, Proposition 4.1] hold because $\omega_0(\alpha)$ agrees exactly with the model metric outside a compact set. Hence there exists a smooth function u such that

$$(\omega_0(\alpha) + i\partial\bar{\partial}u)^3 = e^{f_\alpha} \omega_0(\alpha)^3 = 6i\alpha \Omega \wedge \bar{\Omega}.$$

Since f_α is compactly supported, it satisfies all the polynomially weighted estimates required in [8, Proposition 4.8]. That theorem yields a constant $\bar{u} \in \mathbb{R}$ and some $\delta > 0$ such that

$$|\nabla_{\omega_0(\alpha)}^j i\partial\bar{\partial}u|_{\omega_0(\alpha)} = O(r^{-2-j-\delta})$$

for every $j \geq 0$. Since $\omega_0(\alpha) = \omega_{\text{mod}}$ near infinity, the asserted asymptotic estimate follows. In particular, ω_{CY} is asymptotic to the complete model metric and is therefore complete. $\square$

Proof of Theorem 5.1. Combining the compactifications and uniform model calculations in Subsection 5.3 with Proposition 6.1 and Corollary 6.2 completes the proof of Theorem 5.1. $\square$

7. NONTRIVIALITY OF THE GEOMETRIC MODELS

In this section, we show that the geometric models constructed above are genuinely new in two complementary senses. Recall that the diagonal action defining a threefold of rational-elliptic type (Definition 1.2) is free. The familiar construction on such a quotient is obtained by taking a gravitational instanton on the complement of a fiber in a rational elliptic surface [8], multiplying it by a flat elliptic curve, and descending the invariant product metric. We first prove that the isotrivial quotient models in Cases 5–13 and all the fiber-product models in Theorem 5.1 cannot be recovered in this way. More precisely, after the distinguished fiber is removed, the resulting open threefold is not biholomorphic to the complement of a fiber in any threefold of rational-elliptic type. Second, although the generalized ALH_{b_1, b_2}^* metrics associated with an $I_{b_1}^* \times I_{b_2}^*$ fiber have the same $r^{3/2}$ volume growth as three-dimensional Tian-Yau metrics, their ends have different topology. Consequently, these metrics cannot be recovered from the Tian-Yau construction, even after allowing a change of complex structure analogous to hyperkähler rotation in complex dimension two.

7.1. The monodromy obstruction to rational-elliptic type. We begin by observing that, for the open manifolds considered here, the abelian fibration is determined intrinsically by the complex structure of its total space.

Proposition 7.1. *Let M_1, M_2 be two normal complex spaces, $f_i : M_i \rightarrow \mathbb{C}$ be proper surjective holomorphic maps with connected fibers. Then every biholomorphism $\Phi : M_1 \rightarrow M_2$ induces a unique biholomorphism $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ such that*

$$f_2 \circ \Phi = \sigma \circ f_1.$$

Moreover, in the standard coordinate on $\mathbb{C}$,

$$\sigma(z) = az + b$$

for some $a \in \mathbb{C}^*$ and $b \in \mathbb{C}$.

Proof. By analytic Stein factorization and the normality of $\mathbb{C}$, the assumptions that f_i is proper and has connected fibers imply

$$(f_i)_* \mathcal{O}_{M_i} = \mathcal{O}_{\mathbb{C}}.$$

Since $\mathbb{C}$ is Stein, f_i is therefore the Remmert reduction of M_i . In particular, the holomorphic map $f_2 \circ \Phi : M_1 \rightarrow \mathbb{C}$ factors uniquely through f_1 , giving a holomorphic map $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ such that $f_2 \circ \Phi = \sigma \circ f_1$.

Applying the same argument to Φ^{-1} gives a holomorphic map $\tau : \mathbb{C} \rightarrow \mathbb{C}$ satisfying $f_1 \circ \Phi^{-1} = \tau \circ f_2$. The surjectivity of f_1 and f_2 then implies

$$\tau \circ \sigma = \text{id}_{\mathbb{C}}, \quad \sigma \circ \tau = \text{id}_{\mathbb{C}}.$$

Thus σ is a biholomorphism. Every automorphism of $\mathbb{C}$ is affine, so $\sigma(z) = az + b$ with $a \neq 0$. Uniqueness follows from the surjectivity of f_1 . $\square$

Proposition 7.1 applies to all the examples discussed in the rest of the section, which are all complex manifolds fibered over $\mathbb{C}$ with proper fibration map.

We next introduce a natural invariant that will distinguish the fibrations.

Definition 7.2 (Monodromy fixed subspace). *Let $f : M \rightarrow B$ be a proper holomorphic map and let $U \subset B$ be the locus over which f is smooth. Fix $t \in U$ and denote the monodromy representation by*

$$\rho_f : \pi_1(U, t) \rightarrow \text{GL}(H^1(M_t, \mathbb{Q})).$$

We set

$$H^1(M_t, \mathbb{Q})^{\text{Mon}(f)} := \{v \in H^1(M_t, \mathbb{Q}) : \rho_f(\gamma)v = v \text{ for every } \gamma \in \pi_1(U, t)\}.$$

Equivalently,

$$H^1(M_t, \mathbb{Q})^{\text{Mon}(f)} \cong H^0(U, R^1(f|_{f^{-1}(U)})_* \mathbb{Q}),$$

where the right-hand side is the space of global flat sections of the cohomology local system. Its dimension is independent of the choice of t .

Proposition 7.1 shows that a biholomorphism between two such open total spaces identifies their smooth-fiber loci, up to an affine change of the base coordinate, and identifies their cohomology local systems. Consequently, $\dim_{\mathbb{Q}} H^1(M_t, \mathbb{Q})^{\text{Mon}(f)}$ is an invariant of the open complex manifold in the present setting.

For a threefold of rational-elliptic type, the elliptic curve factor in the construction will provide a 2-dimensional monodromy-invariant space. To be precise, we have the following lemma.

Lemma 7.3. *Let $Y = (S \times E)/\Gamma$ be a threefold of rational-elliptic type, and let $f : Y \rightarrow \mathbb{P}^1$ be its natural abelian-surface fibration. If $p \in \mathbb{P}^1$ and $M = Y \setminus f^{-1}(p)$, then, for a smooth fiber Y_t of $f|_M$,*

$$\dim_{\mathbb{Q}} H^1(M_t, \mathbb{Q})^{\text{Mon}(f)} \geq 2.$$

Proof. Over the smooth locus of the elliptic fibration $g : S \rightarrow \mathbb{P}^1$, the quotient map restricts fiberwise to an isogeny

$$S_t \times E \longrightarrow (S_t \times E)/\Gamma = Y_t.$$

Pullback by these isogenies identifies $R^1(f|_{f^{-1}(U)})_*\mathbb{Q}$ with

$$R^1(g|_{g^{-1}(U)})_*\mathbb{Q} \oplus H^1(E, \mathbb{Q}),$$

where U is the common smooth-fiber locus. Translations act trivially on $H^1(E, \mathbb{Q})$, hence

$$H^1(E, \mathbb{Q}) \subseteq H^1(Y_t, \mathbb{Q})^{\text{Mon}(f)}.$$

Since $\dim_{\mathbb{Q}} H^1(E, \mathbb{Q}) = 2$, the lemma follows. $\square$

We now compute this obstruction in the isotrivial models. Recall that Cases 1–4 in Table 1 have a fixed elliptic factor and were already identified as product-type constructions. We therefore restrict attention to Cases 5–13.

Proposition 7.4 (Nontriviality of the isotrivial quotient models). *For each of the isotrivial quotient models in Cases 5–13, let $f : M \rightarrow \mathbb{C}$ be the abelian fibration obtained after deleting the fiber over infinity. Then M is not biholomorphic to the complement of a fiber in any threefold of rational-elliptic type.*

Proof. The smooth part of the base of f is $\mathbb{C}^*$. Let A denote a smooth abelian-surface fiber, τ denote the monodromy action of a generator of $\pi_1(\mathbb{C}^*)$ on $H^1(A, \mathbb{Q})$.

The eigenvalues of τ on $H^{1,0}(A)$ are as follows:

Case	Eigenvalues of τ on $H^{1,0}(A)$	Case	Eigenvalues of τ on $H^{1,0}(A)$
5	(ζ_3, ζ_3)	10	(i, ζ_3^2)
6	$(i, -1)$	11	(ζ_5, ζ_5^3)
7	$(\zeta_3, -1)$	12	$(\zeta_3^2, -\zeta_3^2)$
8	(ζ_6, ζ_3^2)	13	$(\zeta_6, -\zeta_6)$
9	$(\zeta_6, -i)$		

None of these eigenvalues is equal to 1. The eigenvalues on $H^{0,1}(A)$ are their complex conjugates and are also different from 1. Therefore 1 is not an eigenvalue on $H^1(A, \mathbb{C})$, and hence

$$H^1(A, \mathbb{Q})^{\text{Mon}(f)} = 0. \quad (7.1)$$

Suppose that M_j were biholomorphic to $Y \setminus h^{-1}(q)$ for a threefold Y of rational-elliptic type. By Proposition 7.1, this biholomorphism would identify the two fibrations up to an affine automorphism of $\mathbb{C}$, and hence would identify their monodromy-fixed subspaces. Equation (7.1) would then contradict Lemma 7.3. $\square$

The same invariant also distinguishes all the non-isotrivial fiber-product models.

Proposition 7.5 (Nontriviality of the fiber-product models). *Let $\pi_i : X_i \rightarrow \mathbb{P}^1$, $i = 1, 2$, be the rational elliptic surfaces occurring in Theorem 5.1. Let F be the singular fiber removed in Theorem 5.1 and $M = (X_1 \times_{\mathbb{P}^1} X_2) \setminus F$, and set $p = f(F)$. Then M is not biholomorphic to the complement of a fiber in any threefold of rational-elliptic type.*

Proof. Let $\mathfrak{D}_i \subset \mathbb{P}^1$ be the discriminant of π_i and put $U = \mathbb{P}^1 \setminus (\mathfrak{D}_1 \cup \mathfrak{D}_2)$. Since $p \in \mathfrak{D}_1 \cap \mathfrak{D}_2$ is deleted from the base of M , the set U is precisely the locus over which $f|_M$ is a smooth abelian-surface fibration. For $i = 1, 2$, set $\mathbb{V}_i = R^1(\pi_i|_U)_*\mathbb{Q}$. We first show that $H^0(U, \mathbb{V}_i) = 0$.

Write $\mathfrak{D}_i = \{p, q_{i,1}, \dots, q_{i,r_i}\}$, and denote the corresponding local monodromies on $H^1((X_i)_t, \mathbb{Q})$ by $T_{i,p}$ and $T_{i,j}$. With the usual choice of geometric loops, one has

$$T_{i,p} T_{i,1} \cdots T_{i,r_i} = \text{id}. \quad (7.2)$$

The punctures contributed by the discriminant of the other elliptic surface have trivial monodromy on $\mathbb{V}_i$. Therefore, every vector fixed by the monodromy over U is fixed by all $T_{i,j}$ and, by equation (7.2), is also fixed by $T_{i,p}$.

For the distinguished Kodaira fibers appearing in Theorem 5.1, the characteristic polynomial of $T_{i,p}$ is given by

fiber type	characteristic polynomial
II, II*	$x^2 - x + 1$
III, III*	$x^2 + 1$
IV, IV*	$x^2 + x + 1$
I _b * ($b \geq 0$)	$(x + 1)^2$

In every case, 1 is not an eigenvalue. It follows that

$$H^0(U, \mathbb{V}_i) = 0, \quad i = 1, 2.$$

By the Künneth decomposition, the degree-one cohomology local system of the smooth part of the fiber product is $R^1(f|_{f^{-1}(U)})_* \mathbb{Q} \cong \mathbb{V}_1 \oplus \mathbb{V}_2$. Consequently,

$$H^1(M_t, \mathbb{Q})^{\text{Mon}(f)} = 0. \quad (7.3)$$

If M were biholomorphic to the complement of a fiber in a threefold of rational-elliptic type, Proposition 7.1 would identify the two fibrations and their monodromy local systems. Equation (7.3) would then contradict Lemma 7.3. $\square$

By now, we can finally prove our main Theorem 1.3.

Proof of Theorem 1.3. We collect the results established in the preceding sections.

First, consider the three-dimensional isotrivial quotient models in Cases 5–13. By Theorem 4.1, each such model admits a projective relatively minimal compactification $f : X \rightarrow \mathbb{P}^1$ which is smooth away from the distinguished fiber $F = f^*(\infty)$, has at worst klt singularities along F , and satisfies

$$K_X \sim_{\mathbb{Q}} -\frac{2}{k}F.$$

Moreover, $X \setminus F$ carries a complete Calabi–Yau metric of generalized ALG type. Proposition 7.4 shows that, for Cases 5–13, the open manifold $X \setminus F$ is not biholomorphic to the complement of a fiber in any threefold of rational-elliptic type.

Next, consider the fiber-product models of Theorem 5.1. After applying an automorphism of $\mathbb{P}^1$, we may assume that the distinguished point is ∞ . Theorem 5.1 provides a normal projective compactification $f : X \rightarrow \mathbb{P}^1$ which is smooth away from $F = f^*(\infty)$, has at worst klt singularities, and is relatively minimal. In every complete case appearing in that theorem, one has

$$K_X \sim_{\mathbb{Q}} -\lambda F$$

for some $\lambda \in \mathbb{Q}_{>0}$. The same theorem, together with Proposition 6.1 and Corollary 6.2, yields complete Calabi–Yau metrics of generalized ALG, ALH, ALG*, and ALH* type, with the asymptotic type determined by the local monodromy of the deleted fiber. Finally, Proposition 7.5 shows that none of these open fiber-product models is biholomorphic to the complement of a fiber in a threefold of rational-elliptic type.

Thus all four conditions in Theorem 1.3 are satisfied, and the theorem follows. $\square$

7.2. Comparison with the Tian–Yau construction. We next compare the $I_{b_1}^* \times I_{b_2}^*$ fiber-product models with the Tian–Yau construction. We first recall the relation between the Tian–Yau and ALH* constructions in complex dimension two.

Let D be a compact Kähler manifold with trivial canonical bundle and let $L \rightarrow D$ be an ample line bundle. The Calabi ansatz defines a complete Ricci-flat Kähler metric near the zero section of the punctured line bundle $L \setminus D$. A Calabi–Yau manifold is called asymptotically Calabi if, outside

a compact set, its complex structure, holomorphic volume form, and Kähler form converge to this Calabi model with the prescribed decay. The following theorem shows that such an asymptotic model determines a Tian–Yau compactification and, in complex dimension two, relates the two known constructions of ALH^* gravitational instantons.

The following result identifies the Tian–Yau and ALH^* constructions in complex dimension two.

Theorem 7.6 (Hein–Sun–Viaclovsky–Zhang). [9, Theorem 1.4] *Let $(Z, g, \omega_1, \omega_2, \omega_3)$ be an ALH^* gravitational instanton with parameter b . In the complex structure I determined by ω_1 , the complex surface (Z, I) is biholomorphic to $\bar{Z} \setminus D$, where $\bar{Z}$ is a weak del Pezzo surface and D is a smooth anticanonical elliptic curve with $D^2 = b$; under this identification, g is a generalized Tian–Yau metric. After hyperkähler rotation to a second complex structure J , the same Riemannian manifold is biholomorphic to $S \setminus F$, where S is a rational elliptic surface with a section and F is a multiplicity-one fiber of type I_b .*

Thus, in complex dimension two, the Tian–Yau and ALH^* constructions give two complex-geometric descriptions of the same complete hyperkähler manifold. The compactifications $\bar{Z}$ and S are generally different; their relationship is established through hyperkähler rotation rather than through an isomorphism of the compactified surfaces. This surface correspondence motivates the higher-dimensional question considered below.

In complex dimension n , a Calabi-model metric has volume growth of order $r^{2n/(n+1)}$. In particular, a three-dimensional Tian–Yau metric has volume growth of order $r^{3/2}$. This is exactly the volume-growth order of the generalized ALH_{b_1, b_2}^* metrics constructed from an $I_{b_1}^* \times I_{b_2}^*$ fiber. The surface theorem therefore suggests the following naive higher-dimensional analogue.

Remark 7.7. *Can a generalized ALH_{b_1, b_2}^* metric arising from an $I_{b_1}^* \times I_{b_2}^*$ fiber be identified with a three-dimensional Tian–Yau metric on the complement of a smooth anticanonical Calabi–Yau surface in a weak Fano threefold, possibly after changing the complex structure in analogy with hyperkähler rotation?*

The answer is negative for a purely topological reason: The end of the Tian–Yau model is homotopic to a circle bundle over a K3 surface, but the end of the generalized ALH_{b_1, b_2}^* metric is homotopic to a mapping torus of certain monodromy. We can distinguish these two ends by explicit topological invariant.

Proposition 7.8. *Let M_{b_1, b_2} be the complement associated with an $I_{b_1}^* \times I_{b_2}^*$ fiber in Theorem 5.1. Then the end of M_{b_1, b_2} is not diffeomorphic to the end of any complex three-dimensional asymptotically Calabi manifold. Consequently, its generalized ALH_{b_1, b_2}^* Calabi–Yau metric cannot be identified with a Tian–Yau metric, even after changing the complex structure.*

Proof. A cross-section N_{b_1, b_2} of the generalized ALH_{b_1, b_2}^* end is the mapping torus of the monodromy

$$A = A_{b_1} \oplus A_{b_2} : T^4 \longrightarrow T^4, \quad A_b = - \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix},$$

up to replacing the matrices by their inverses or transposes. Since $\det(A_b - \text{id}) = 4$, the map $A - \text{id}$ is invertible over $\mathbb{Q}$. Set $C_A := \frac{H_1(T^4, \mathbb{Q})}{(A - \text{id})H_1(T^4, \mathbb{Q})}$. The Wang exact sequence of the mapping torus gives

$$0 \longrightarrow C_A \longrightarrow H_1(N_{b_1, b_2}, \mathbb{Q}) \longrightarrow H_0(T^4, \mathbb{Q}) \longrightarrow 0.$$

The cokernel in this sequence vanishes, and therefore

$$H_1(N_{b_1, b_2}, \mathbb{Q}) \cong \mathbb{Q}, \quad b_1(N_{b_1, b_2}) = 1. \quad (7.4)$$

On the other hand, a cross-section of a complex three-dimensional Calabi end is the unit-circle bundle

$$S^1 \longrightarrow P(L) \longrightarrow D,$$

where D is a compact projective surface with $K_D \cong \mathcal{O}_D$ and $L \rightarrow D$ is ample. The relevant part of the Gysin sequence is

$$0 \longrightarrow H^1(D, \mathbb{Q}) \longrightarrow H^1(P(L), \mathbb{Q}) \longrightarrow H^0(D, \mathbb{Q}) \xrightarrow{\sim_{c_1(L)}} H^2(D, \mathbb{Q}).$$

Since L is ample, $c_1(L) \neq 0$, so the last map is injective. Hence

$$H^1(P(L), \mathbb{Q}) \cong H^1(D, \mathbb{Q}). \quad (7.5)$$

Thus $b_1(P(L)) = 0$, since D is a K3 surface. Hence equation (7.5) is incompatible with equation (7.4).

Both ends are products of their respective cross-sections with a half-line outside a compact set. A diffeomorphism between the ends would therefore induce an isomorphism of their first rational cohomology groups, which is impossible. Since changing a complex structure does not change the topology of the underlying end, the claimed non-equivalence follows. $\square$

Remark 7.9. *The common $r^{3/2}$ volume growth conceals different anisotropic geometries. On a three-dimensional Calabi end, the four directions tangent to D have length scale $r^{1/4}$, while the circle-fiber direction has length scale $r^{-1/2}$. On the generalized ALH_{b_1, b_2}^* end, one angular direction has length scale $r^{1/2}$, two torus directions have length scale $r^{1/4}$, and two torus directions have length scale $r^{-1/4}$. In both cases the cross-sectional volume is of order $r^{1/2}$, leading after radial integration to volume growth of order $r^{3/2}$. The coincidence of the volume-growth exponent therefore does not reflect an equivalence of the underlying models.*

Propositions 7.4, 7.5, and 7.8 show that the constructions in this paper are not disguised forms of the previously known product-quotient or Tian–Yau constructions. In particular, the $I_{b_1}^* \times I_{b_2}^*$ construction produces a genuinely new geometric model and a new class of complete Calabi–Yau metrics in complex dimension three.

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